Magic Gems: A Polyhedral Framework for Magic Squares

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Abstract

To an $n \times n$ magic square we attach the n^2 points (x, y, z) in $\mathbb{R}^3$, where (x, y) is the position of a cell measured from the centre of the grid and z is its entry minus the mean entry; their convex hull is the *Magic Gem*. Read as vectors from their centroid, the points balance for every arrangement of the entries, whereas the heights balance as weights on the grid exactly when the rows and columns have equal sums: the row and column conditions are equivalent to the vanishing of all height-weighted moments along the two axes, among them the dipole moment, while the mixed moment is not forced. Consequently the covariance matrix of the points is the same for all semi-magic squares of a given order, and a short argument shows that the point set has a non-trivial isometry exactly when the square equals its complement up to rotation and reflection. The lower and upper faces of a gem are the regular subdivisions of the grid induced by the entries and by their negatives. For the 880 magic squares of order four we compute all gems exactly, by computer: no gem has an isometry that its point set lacks; 48 gems are centrally symmetric, 304 have a single half-turn symmetry and 528 have none; the vertex set determines the square; and the gems form 616 congruence classes and 352 combinatorial types. For larger orders the vertex count can grow: other linear magic squares have up to 18, 25, 31 and 36 vertices for orders 5, 7, 9 and 11. For the Siamese (de la Loubère) square of every odd order $n \geq 5$, however, we prove that the gem has exactly the twelve vertices that the boundary of the grid forces (the four corners and, on each side, the cells of the smallest and largest entry), with f-vector $(12, 26, 16)$ and isometry group $\{\pm\mathrm{id}\}$; the proof reduces to checking the signs of a table of polynomials in $m = (n-1)/2$.

Keywords: magic squares, convex polytopes, regular subdivisions, symmetry groups, Siamese method
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1 Introduction

Magic squares are among the oldest objects of recreational mathematics: the 3×3 Lo Shu belongs to Chinese tradition, a 4×4 square is engraved in Dürer's *Melencolia I* (1514), and constructions for every order $n \geq 3$ have been known for centuries [2, 25]. The modern literature treats them from several directions: enumeration and the geometry of the polytopes whose lattice points are the squares [31, 12, 1, 6, 5], spectral and matrix properties [19, 22, 23, 7], and, in work of Loly and collaborators, physical invariants such as the moment of inertia and the multipole moments [17, 27, 28].

Definition 1.1 (Magic square)**.** An *arrangement* of order n is a bijection $S = (s_{ij})$ from the cells of the $n \times n$ grid onto $\{1, \ldots, n^2\}$ (a permutation of $1, \ldots, n^2$ written into the grid). It is a *magic square* if every row, every column and both main diagonals sum to the magic constant $\mu_n = n(n^2+1)/2$, and *semi-magic* if only the rows and columns do.

The construction. Centre the grid at the origin and attach to the cell (i, j) the point

$$\mathbf{p}_{ij} = (x_{ij}, y_{ij}, z_{ij}),$$

whose horizontal part (x_{ij}, y_{ij}) is the position of the cell and whose height $z_{ij} = s_{ij} - \frac{n^2+1}{2}$ is the entry minus the mean entry. Read as vectors from the origin, the $\mathbf{p}_{ij}$ of a magic square have vertical components that cancel along every row, column and both diagonals (Lemma 2.2; Figures 1 and 2). The *Magic Gem* $\mathcal{G}(S)$ is the convex hull of the n^2 points (Section 2). Figure 1 shows Dürer's square, its vectors and its gem, which has 12 vertices and 4 points in its interior.

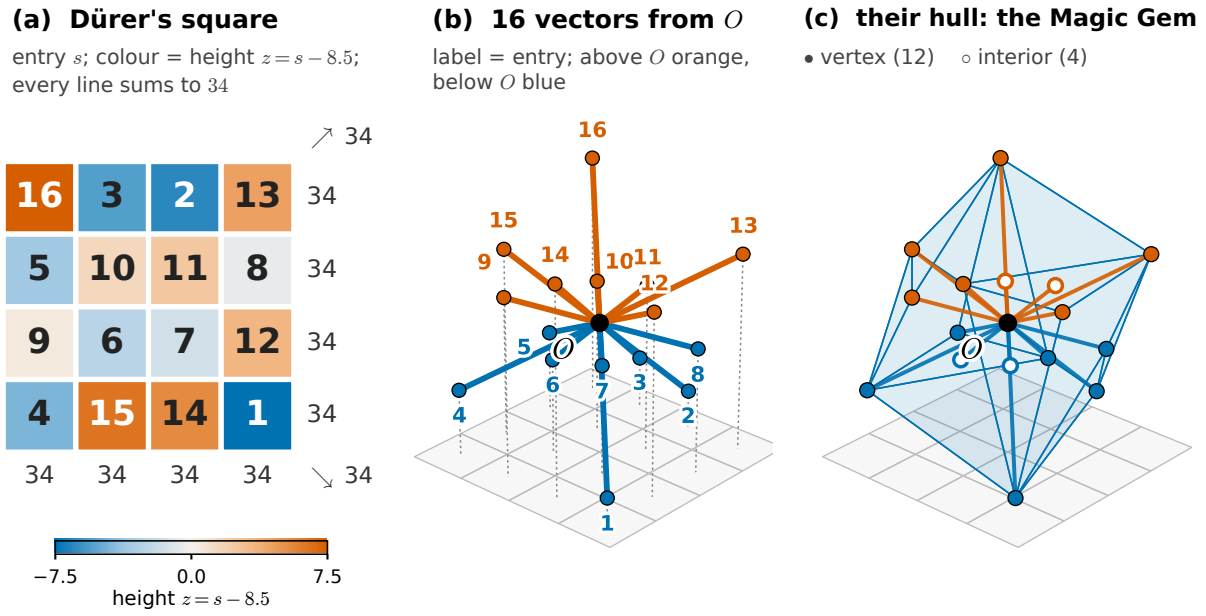


Figure 1: Dürer's square. (a) Entries s_{ij} on a heat map of the heights $z_{ij} = s_{ij} - 8.5$ (orange above 0, blue below); every line sums to 34. (b) The 16 vectors $\mathbf{p}_{ij}$ from the origin O, labelled by the entries, with drop lines to the floor grid. (c) Their convex hull, the Magic Gem: 12 tips are vertices (filled), 4 lie in the interior (open). The colour bar is shared by (a) and (b).

Results. The paper proceeds from what the pictures show to what can be proved about them, and from the simplest objects to the most rigid ones. The vectors $\mathbf{p}_{ij}$ balance about the origin for every arrangement; what distinguishes magic squares is the balance of the heights as weights on the grid (item 1). This balance fixes the covariance matrix of the point set, which settles the symmetries and congruences of the point sets for every order (item 2). The convex hull keeps only the extreme points, and the remaining results ask how much of the square the hull still sees. At order four it has no isometry that the point set lacks, and its vertices alone determine the square (item 3). For the Siamese squares of every odd order $n \geq 5$ it sees only the twelve entries that the boundary of the grid forces (item 4).

1. *Balance* (Section 3). For every arrangement the origin is the centroid of the points. An arrangement is semi-magic if and only if $\sum z_{ij}\varphi(x_{ij}) = \sum z_{ij}\varphi(y_{ij}) = 0$ for every function φ; in particular the dipole moment $\sum z\,(x, y)$ vanishes, while the mixed moment $\sum z\,xy$ is not forced. The covariance matrix of the $N = n^2$ points of a semi-magic square is $\mathrm{diag}(a_n, a_n, b_n)$ with $a_n = (n^2 - 1)/12 \neq b_n = (n^4 - 1)/12$ (Lemma 3.2).

2. *Symmetry and congruence for every order* (Theorem 4.2). Every isometry of the point set of a semi-magic square lies in $D_4 \times \{\pm 1\}$, the point set has a non-trivial isometry if and only if S is self-complementary, and two semi-magic squares have congruent point sets if and only if one is a D_4-image of the other or of its complement.

3. *The gems of order four* (Theorem 4.3, Section 5). There are 880 magic squares of order four up to rotation and reflection (D_4-classes). The isometry group of each gem equals that of its point set; so 48 gems (the associative squares) are centrally symmetric, 304 have one half-turn, and 528 have no symmetry. The lower and upper faces of a gem are the regular subdivisions

of the grid induced by z and $-z$ (Proposition 5.2). We tabulate the exact geometry (8 to 16 vertices, 39 f-vectors, 352 combinatorial types, 155 distinct volumes, 616 congruence classes) and show that the vertex set alone determines the square up to D_4 (Proposition 5.4).

4. *Siamese gems* (Theorem 6.2). For the Siamese square of every odd order $n \geq 5$ the vertices of the gem are exactly the twelve vertices that the boundary of the grid forces (Lemma 5.1), the f-vector is $(12, 26, 16)$, and the isometry group is $\{\pm\mathrm{id}\}$. The proof is complete on paper: a three-case reduction, a table of 128 polynomial slacks (a slack is the right-hand side minus the left-hand side of a facet inequality, evaluated at a point), and a closing argument on 16 facets. Other linear magic squares can have many more vertices: up to 18, 25, 31 and 36 for $n = 5, 7, 9, 11$, and as few as 11 at order five, while some order-five magic squares that are not linear have only 10 (Remark 6.3).

Relation to previous work. The polyhedral geometry of magic squares is classical: nonnegative integer matrices with all line sums equal to t are the lattice points of the t-th dilate of the Birkhoff polytope, and their number is a polynomial in t [31, 12, 34]; the count of magic squares with distinct entries is an Ehrhart quasi-polynomial of an inside-out polytope [6, 5]; the cones of magic squares and cubes have Hilbert-basis descriptions [1]; contingency tables with fixed margins are the same lattice points seen as a sample space [10]. In all of these the magic squares are the *points* of a polytope, whereas the Magic Gem attaches a polytope to a *single* square. The lifting construction itself is that of regular subdivisions and secondary fans [15, 9]. The moment identities of Section 3 are due in substance to Loly and Rogers [17, 27, 28]; we record them in the coordinates of the gem for later use. The partition of the 880 order-four squares into associative, self-complementary and other squares is Dudeney's [11]. New are the Siamese theorem and the exact order-four census, together with a short covariance argument that settles symmetry and congruence of the point sets for every order. The moment identities of all orders implied by the row and column sums, the number of them that forces an arrangement to be semi-magic (a permutation threshold), and the line-sum energies of random arrangements are treated in a companion manuscript [20]; no result of it is used here.

Section 2 defines the gem and describes what the figures show, Section 3 the balance of the vectors and the moment identities, Section 4 symmetry and congruence, Section 5 the forced vertices, the regular subdivisions and the order-four census, Section 6 the Siamese gems, and Section 7 lists open questions. Appendix A describes the computations, and Appendix B records the Ehrhart polynomials of the order-four gems and numerical vertex counts for larger orders. An interactive display of the 880 order-four gems is available at `https://kylemath.github.io/MagicGemWebpage/`.

2 The Magic Gem

Let $S = (s_{ij})$ be an arrangement of order $n \geq 2$, with cells indexed by $i, j \in \{0, \ldots, n-1\}$ (i the row from the top, j the column from the left), and write $N = n^2$ and $\bar{s} = (N+1)/2$ for the mean entry.

Definition 2.1 (Lifted point set and Magic Gem)**.** The Magic Gem is the convex hull of the lift of the grid by the heights z defined below (the lifting used for regular subdivisions). To the cell (i, j) assign the point $\mathbf{p}_{ij} = (x_{ij}, y_{ij}, z_{ij}) \in \mathbb{R}^3$ with

$$x_{ij} = j - \frac{n-1}{2}, \qquad y_{ij} = \frac{n-1}{2} - i, \qquad z_{ij} = s_{ij} - \bar{s}. \tag{1}$$

The *Magic Gem* of S is the convex hull $\mathcal{G}(S) = \mathrm{conv}\{\mathbf{p}_{ij}\}$, and $z = (z_{ij})$ is the *height function* of S.

The x-axis runs along the columns and the y-axis along the rows (top row at positive y). Let $x_j = j - \frac{n-1}{2}$ and $y_i = \frac{n-1}{2} - i$ denote the n values taken by x and y. A *line* is one of the n rows,

the n columns, the main diagonal $\{i=j\}$ or the anti-diagonal $\{i+j=n-1\}$. For a line L let $\delta_L(S)=\sum_{c\in L}s_c-\mu_n$ be the *excess* of the line sum over the magic constant, so S is magic iff $\delta_L(S)=0$ for all $2n+2$ lines. Moment and centroid statements below refer to the N points $\mathbf{p}_{ij}$ with unit weights, not to the solid $\mathcal{G}(S)$.

Terminology. Each point $\mathbf{p}_{ij}$ is either a *vertex* of $\mathcal{G}(S)$, an *interior point* (in the interior of $\mathcal{G}(S)$), or a *non-vertex boundary point* (on the boundary but not a vertex). We write $\operatorname{vert}\mathcal{G}(S)$ for the vertex set, V, V°, V^∂ for the three numbers ($V+V^\circ+V^\partial=N$), and (V,E,F) for the f-vector. For any set $X\subseteq\mathbb{R}^3$, $\mathrm{Iso}(X)$ is the group of isometries of $\mathbb{R}^3$ that map X onto itself. The dihedral group $D_4=\langle r,h\rangle$ of the square acts on arrangements by $(gS)_{gc}=s_c$; here r is the quarter-turn, r^2 the half-turn, h and v the left–right and top–bottom reflections and d the reflection in the main diagonal $i=j$. For $g\in D_4$ let $\hat{g}(x,y,z)=(g(x,y),z)$, where g acts on the plane as the corresponding orthogonal map. The order-four squares are numbered $\#0,\dots,\#879$ in the lexicographic order of their Frénicle standard forms (the image under D_4 with the smallest corner entry at top left and $s_{01}<s_{10}$) [13]; the list is `anc/squares880.txt`.

Lemma 2.2 (Means, variances, line sums)**.** *For every arrangement S of order n:*

1. $\sum_{i,j}x_{ij}=\sum_{i,j}y_{ij}=\sum_{i,j}z_{ij}=0$ *and* $\sum_{i,j}x_{ij}y_{ij}=0$*;*
2. $\frac{1}{N}\sum x_{ij}^2=\frac{1}{N}\sum y_{ij}^2=a_n:=\frac{n^2-1}{12}$ *and* $\frac{1}{N}\sum z_{ij}^2=b_n:=\frac{N^2-1}{12}$*;*
3. *for every line L, $\delta_L(S)=\sum_{c\in L}z_c$, and $\sum_{c\in L}\mathbf{p}_c=\big(n\,\bar{x}_L,\ n\,\bar{y}_L,\ \delta_L(S)\big)$, where $(\bar{x}_L,\bar{y}_L)$ is the centre of the line: $(0,y_i)$ for the row i, $(x_j,0)$ for the column j, and $(0,0)$ for both diagonals.*

In particular S is magic if and only if the vector sum of the $\mathbf{p}_c$ along every line is horizontal.

Proof. (1) $\sum_{i,j}x_{ij}=n\sum_j x_j=0$, likewise for y; $\sum z=\sum_{k=1}^{N}k-N\bar{s}=0$; and $\sum xy=(\sum_j x_j)(\sum_i y_i)=0$. (2) $\frac{1}{n}\sum_j x_j^2$ is the variance of n equally spaced points with unit spacing, $(n^2-1)/12$; the same holds for y, and for z with N points. (3) Since $\mu_n=n\bar{s}$, $\sum_{c\in L}z_c=\delta_L$. The cells of row i have $y=y_i$ and x running over all x_j; for a column the roles are exchanged; each diagonal is invariant under $(x,y)\mapsto(-x,-y)$. □

For $g\in D_4$ the points of gS are $\hat{g}\,\mathbf{p}_c$, so

$$\mathcal{G}(gS)=\hat{g}\,\mathcal{G}(S),\tag{2}$$

and since D_4 maps magic squares to magic squares, every D_4-class of magic squares has a single gem up to congruence.

Proposition 2.3 (Dimension)**.** *Let S be an arrangement of order $n\geq 2$. The points $\mathbf{p}_{ij}$ are coplanar if and only if S is one of the eight D_4-images of the natural square $s_{ij}=ni+j+1$. Otherwise $\mathcal{G}(S)$ is a 3-polytope with the origin in its interior. In particular this holds for every semi-magic square.*

Proof. The grid positions are not collinear, so a plane containing all points is not vertical and the points are coplanar if and only if $z=\alpha x+\beta y+\gamma$ on the grid. Then $u=\alpha$ and $w=-\beta$ are the differences of the entries of horizontally and vertically adjacent cells, so they are integers, and $s_{ij}=s_{00}+uj+wi$. Since the entries are distinct, $u,w\neq 0$; applying h, v and d we may assume $1\leq u\leq w$, so that $s_{00}=1$ is the smallest entry. The entry 2 is $1+uj+wi$ with $(i,j)\neq(0,0)$, hence $u=1$, and row 0 holds $1,\dots,n$. The entry $n+1$ lies in a row $i\geq 1$, so $w\leq n$; if $w<n$ then the cells $(0,w)$ and $(1,0)$ both hold $1+w$. Hence $w=n$ and $s_{ij}=ni+j+1$; conversely, every D_4-image of this arrangement has affine heights. If the points span $\mathbb{R}^3$ affinely, their centroid, which is $\mathbf{0}$ by Lemma 2.2(1), is a convex combination with positive weights of all of them and hence an interior point of the hull. Finally, if S is semi-magic and $z=\alpha x+\beta y+\gamma$, then $\gamma=0$ by Lemma 2.2(1), the row i has height sum $n\beta y_i=0$ for all i and the column j has $n\alpha x_j=0$, so $\alpha=\beta=0$, which is impossible. □

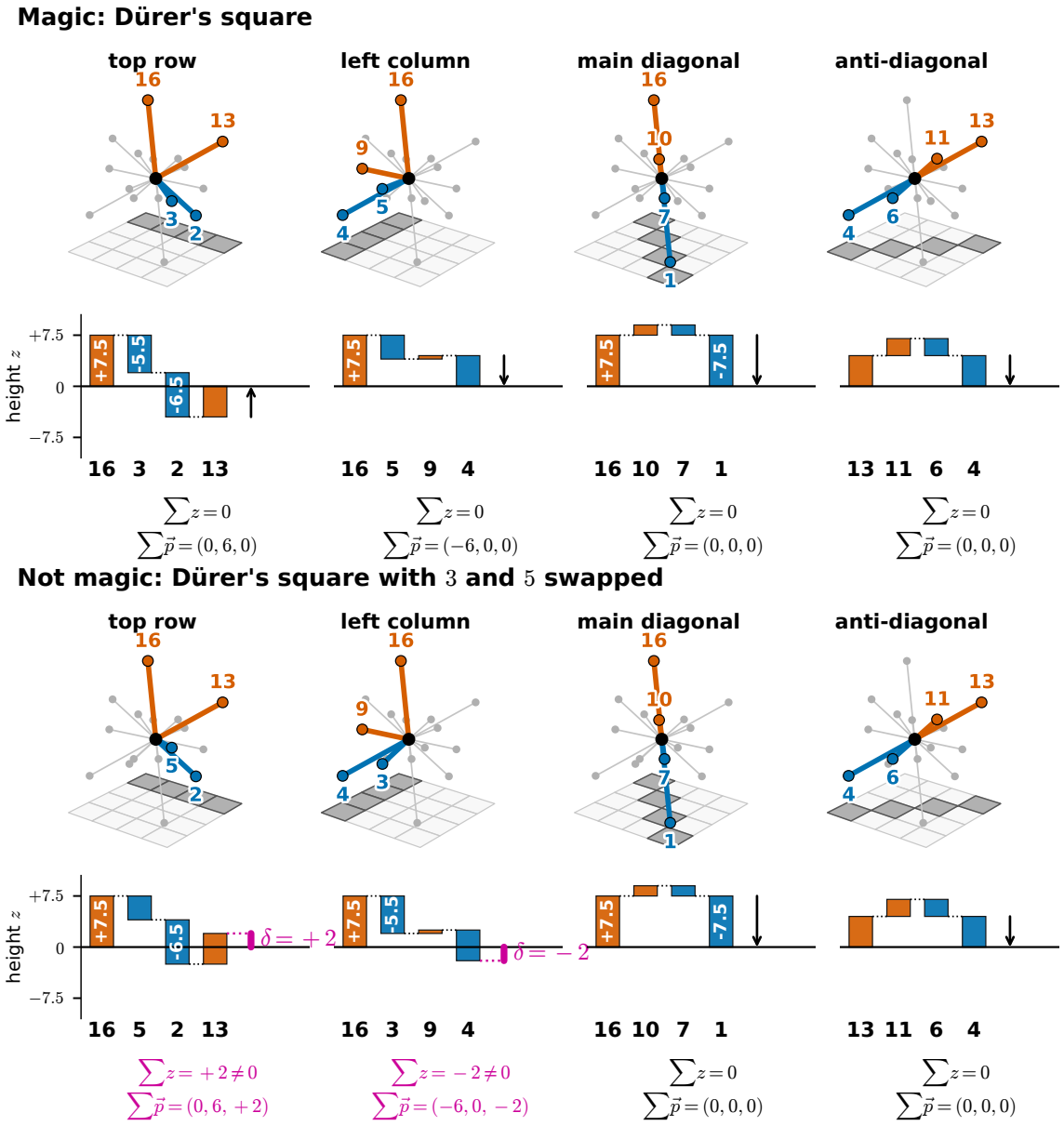


Figure 2: Heights cancel along every line. *Top:* for Dürer's square, the vectors of the top row, left column and both diagonals (coloured; the cells of the line shaded on the floor), and below each the heights $z = s - 8.5$ of the line stacked head to tail; the running total returns to 0, so $\sum_{c \in L} \mathbf{p}_c$ is horizontal (Lemma 2.2(3)). *Bottom:* after swapping the entries 3 and 5 the top row and the left column no longer close (excesses $\delta_L = \pm 2$, magenta); the diagonals do not contain the swapped cells.

The natural square of order 3 shows that the hypothesis is needed: its gem is a parallelogram.

Example 2.4 (Lo Shu)**.** For the Lo Shu square $S = \begin{pmatrix} 2 & 7 & 6 \\ 9 & 5 & 1 \\ 4 & 3 & 8 \end{pmatrix}$, $\bar{s} = 5$ and the nine points are

$$\begin{aligned} \mathbf{p}_{00} &= (-1, 1, -3), & \mathbf{p}_{01} &= (0, 1, 2), & \mathbf{p}_{02} &= (1, 1, 1), \\ \mathbf{p}_{10} &= (-1, 0, 4), & \mathbf{p}_{11} &= (0, 0, 0), & \mathbf{p}_{12} &= (1, 0, -4), \\ \mathbf{p}_{20} &= (-1, -1, -1), & \mathbf{p}_{21} &= (0, -1, -2), & \mathbf{p}_{22} &= (1, -1, 3). \end{aligned}$$

The origin is the only non-vertex point and is interior; the gem has f-vector $(8, 16, 10)$ and volume 18 (Figure 3).

The figures of this section show two things at once: the vectors $\mathbf{p}_{ij}$ balance about the origin, and their heights cancel along every line. Only the second is a property of magic squares. Sec-

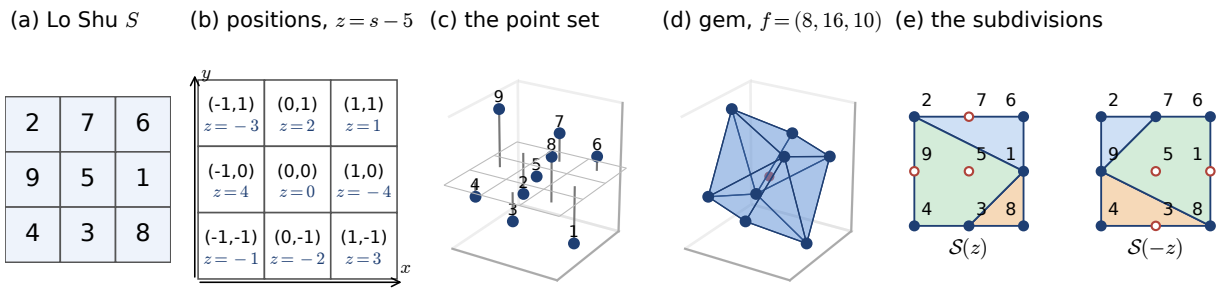


Figure 3: The gem of the Lo Shu square: (a) the square; (b) the cell positions (x_{ij}, y_{ij}), measured from the centre, with the heights $z_{ij} = s_{ij} - 5$; (c) the nine points, labelled by the entries; (d) their convex hull, with f-vector $(8, 16, 10)$ and the interior point $\mathbf{0}$; (e) the regular subdivisions $\mathcal{S}(z)$ and $\mathcal{S}(-z)$ of the grid, the projections of the lower and the upper facets (Example 5.3); grid points that belong to no cell are drawn as open circles.

tion 3 separates the balance statements that hold for every arrangement from those that the line conditions force, and these moment identities are what the symmetry results of Section 4 rest on.

3 Balance: the vectors from the centroid

Figures 1 and 2 draw the points $\mathbf{p}_{ij}$ as vectors from the origin. This section asks which balance statements these vectors satisfy and which of them depend on S being magic. There are two kinds: the vectors themselves balance for every arrangement, while the balance of the heights as weights on the grid is equivalent to the row and column conditions.

Unweighted balance holds for every arrangement. By Lemma 2.2(1), $\sum_{i,j} \mathbf{p}_{ij} = \mathbf{0}$ for every arrangement: the origin is the centroid of the N points, and by Proposition 2.3 it is an interior point of $\mathcal{G}(S)$ unless S is a D_4-image of the natural square. This is a consequence of centring the coordinates and carries no information about S; it does not distinguish magic squares from other arrangements. It is also a statement about the points with unit weights and not about the solid: the centre of mass of the solid $\mathcal{G}(S)$ need not be the origin, and for 829 of the 880 order-four gems it is not (`anc/solid_centroid.py`).

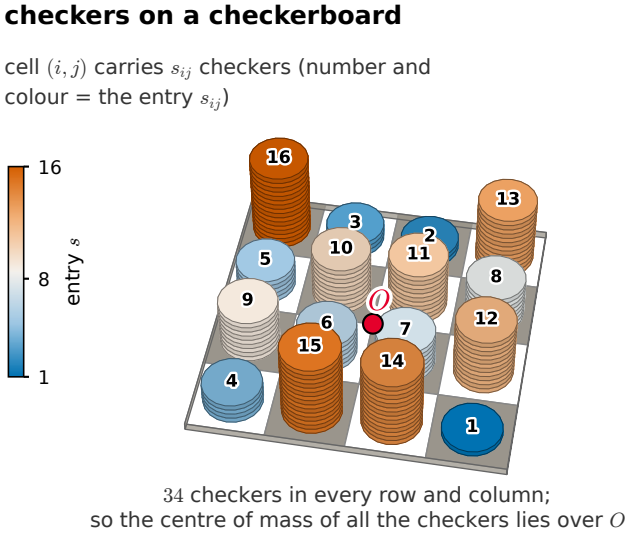


Figure 4: Dürer's square as piles of checkers, s_{ij} on cell (i, j), seen from the front of the board in the orientation of the square; each row and column carries 34 checkers, so the centre of mass of all the checkers lies over the centre O of the board.

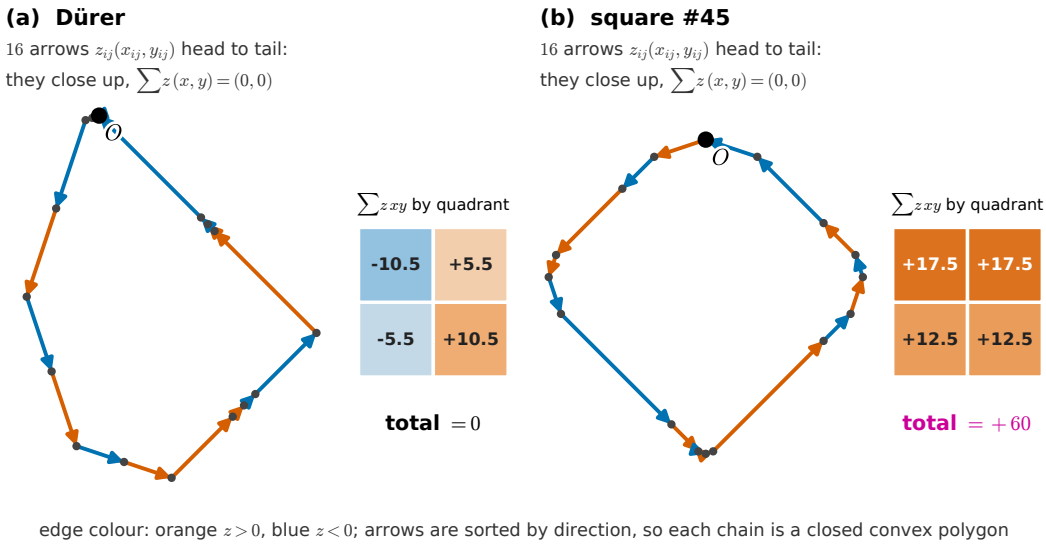


Figure 5: The dipole moment $\sum z_{ij}\,(x_{ij}, y_{ij})$ for Dürer's square and square #45: the 16 vectors $z_{ij}(x_{ij}, y_{ij})$ (orange for $z > 0$, blue for $z < 0$), sorted by direction and laid head to tail, close up at O. The tables give the mixed moment $\sum z\,xy$ over the four quadrants of the grid; the totals are 0 and 60.

Height-weighted balance is the row and column conditions. Now give each cell (x_{ij}, y_{ij}) of the grid the signed weight z_{ij}; the total weight is 0. Write $R_i = \sum_j z_{ij}$ and $C_j = \sum_i z_{ij}$ for the *row and column excesses*; by Lemma 2.2(3), S is semi-magic if and only if all R_i and C_j vanish. For every function $\varphi\colon \mathbb{R} \to \mathbb{R}$, grouping the cells by column and by row gives

$$\sum_{i,j} z_{ij}\,\varphi(x_{ij}) = \sum_j \varphi(x_j)\,C_j, \qquad \sum_{i,j} z_{ij}\,\varphi(y_{ij}) = \sum_i \varphi(y_i)\,R_i. \tag{3}$$

Since the functions $\varphi(x)$ on the grid span the indicator functions of the columns, S is semi-magic if and only if both sums vanish for every φ. For $\varphi(t) = t$ this says that the first moments $\sum z_{ij}x_{ij}$ and $\sum z_{ij}y_{ij}$ of the heights vanish; in the multipole language of Rogers and Loly [28] the *dipole moment* $\sum z\,(x, y)$ of the signed weights is zero. Figure 4 shows this with the raw entries: since $\sum x_{ij} = \sum y_{ij} = 0$, the dipole moment equals $\sum s_{ij}(x_{ij}, y_{ij})$, the first moment of a pile of s_{ij} checkers on each cell, and it vanishes because every row and every column carries the same number of checkers. Figure 5 shows it in vector form: the sixteen vectors $z_{ij}(x_{ij}, y_{ij})$, laid head to tail, close up. For $\varphi(t) = t^2$, (3) gives Loly's invariant.

Corollary 3.1 (Loly [17])**.** *For every arrangement,* $J(S) := \sum_{i,j} s_{ij}(x_{ij}^2 + y_{ij}^2) = \sum_j x_j^2 C_j + \sum_i y_i^2 R_i + \frac{n^2(n^4-1)}{12}$*; in particular* $J(S) = n^2(n^4-1)/12$ *for every semi-magic square.*

Proof. Substitute $s_{ij} = z_{ij} + \bar{s}$, apply (3) with $\varphi(t) = t^2$ to the first part and Lemma 2.2(2) to $\bar{s}\sum(x^2 + y^2) = \bar{s} \cdot 2Na_n$. □

The mixed moment is not forced. The diagonal conditions are not of the form (3), and the line conditions do not determine the mixed moment $\sum z\,xy$. Over the 880 order-four magic squares, $|\sum z\,xy|$ takes 48 distinct values between 0 and 60, and it vanishes for 56 classes (tables in Figure 5). It vanishes whenever S is associative ($s_{ij} + s_{n-1-i,\,n-1-j} = N+1$ for every cell), since then z is odd under $(x, y) \mapsto (-x, -y)$ (Section 4) while xy is even; this accounts for 48 of the 56.

Second moments. The second moments of the point set are as rigid as the first: for a semi-magic square they do not depend on the square at all.

Lemma 3.2 (Covariance)**.** *The covariance matrix* $\Sigma = \frac{1}{N}\sum_{i,j} \mathbf{p}_{ij}\mathbf{p}_{ij}^{\top}$ *of the points of an arrangement* S *has* $\Sigma_{xx} = \Sigma_{yy} = a_n$, $\Sigma_{zz} = b_n$, $\Sigma_{xy} = 0$, $N\Sigma_{xz} = \sum_j x_j C_j$ *and* $N\Sigma_{yz} = \sum_i y_i R_i$. *If* S *is semi-magic,* $\Sigma = \operatorname{diag}(a_n, a_n, b_n)$, *and* $a_n < b_n$ *for* $n \geq 2$.

Proof. Lemma 2.2(1)–(2) and (3) with $\varphi(t) = t$. □

So the first and second moments of the point set are the same for all semi-magic squares of order n. Section 4 turns this rigidity into a description of the symmetries of the point set.

4 Symmetry and congruence

By Lemma 3.2 the point set of every semi-magic square of order n has the same covariance matrix $\operatorname{diag}(a_n, a_n, b_n)$, with a horizontal and a vertical eigenspace. An isometry between two such point sets must preserve these eigenspaces, and this reduces every question about symmetry and congruence of the point sets to the dihedral group of the grid and to complementation (Theorem 4.2). Every symmetry of the point set is a symmetry of its gem, and at order four we show by exact computation that the gems have no others (Theorem 4.3).

4.1 Point sets of every order

Let $\tau(x, y, z) = (x, y, -z)$ be the reflection in the plane $z = 0$. The *complement* of an arrangement is $\overline{S} = N + 1 - S$, with height function $-z$; S is *self-complementary* if $\overline{S} = gS$ for some $g \in D_4$, and *associative* if $\overline{S} = r^2 S$, i.e. if entries in cells symmetric about the centre add up to $N + 1$. Complements of magic squares are magic. Since the points of $\overline{S}$ are $\tau \mathbf{p}_c$,

$$\mathcal{G}(\overline{S}) = \tau\, \mathcal{G}(S). \tag{4}$$

Lemma 4.1 (Elements of D_4 that can complement)**.** *Let $\overline{S} = gS$ for an arrangement S of order $n \geq 2$ and some $g \in D_4$. Then g is unique, $g^2 = \mathrm{id}$, and g fixes no cell except possibly the centre cell of an odd grid, whose entry is then $\bar{s}$. Consequently $g = r^2$ (so S is associative) if n is odd, and $g \in \{r^2, h, v\}$ if n is even. The map $\iota_g = \hat{g}\,\tau$ is an isometry of $\mathbb{R}^3$ that maps the point set of S onto itself, so $\iota_g \in \mathrm{Iso}(\mathcal{G}(S))$; explicitly $\iota_{r^2} = -\mathrm{id}$ (a point reflection), $\iota_h(x, y, z) = (-x, y, -z)$ and $\iota_v(x, y, z) = (x, -y, -z)$ (half-turns about the y- and the x-axis).*

Proof. If also $\overline{S} = g'S$ then $g^{-1}g'$ fixes S; as the entries are distinct it fixes every cell, and D_4 acts faithfully on the cells, so $g = g'$. Since complementation commutes with g, $g^2 S = g\overline{S} = \overline{gS} = \overline{\overline{S}} = S$, so $g^2 = \mathrm{id}$. If g fixes a cell c then $s_c = N + 1 - s_c = 2\bar{s}$. The involutions of D_4 are r^2, h, v and the two diagonal reflections; the diagonal reflections fix $n \geq 2$ cells, h and v fix the n cells of the middle column or row when n is odd and none when n is even, and r^2 fixes only the centre cell when n is odd. Finally, the points of S are mapped by τ to those of $\overline{S} = gS$, and by $\hat{g} = \hat{g}^{-1}$ back to those of $g^2 S = S$, by (2). □

Theorem 4.2 (Point sets of semi-magic squares)**.** *Let S and T be arrangements of order $n \geq 2$ with $\sum z_{ij}x_{ij} = \sum z_{ij}y_{ij} = 0$ (the first moments of the heights vanish), for instance semi-magic squares, and let P_S, P_T be their point sets.*

1. *Every isometry ψ of $\mathbb{R}^3$ with $\psi(P_S) = P_T$ is of the form $\hat{g}$ or $\hat{g}\,\tau$ with $g \in D_4$; $\hat{g}(P_S) = P_T$ if and only if $T = gS$, and $\hat{g}\,\tau(P_S) = P_T$ if and only if $T = g\overline{S}$.*
2. *P_S and P_T are congruent if and only if $T \in D_4 S \cup D_4 \overline{S}$.*
3. *$\mathrm{Iso}(P_S) = \{\mathrm{id}\}$ unless S is self-complementary, and $\mathrm{Iso}(P_S) = \{\mathrm{id}, \iota_g\}$ if $\overline{S} = gS$.*

Proof. An isometry of $\mathbb{R}^3$ mapping one finite set onto another maps centroid to centroid, and both centroids are $\mathbf{0}$ (Lemma 2.2(1)); so ψ is orthogonal, and $\psi \Sigma_S \psi^\top = \Sigma_T$ for the covariance matrices. By Lemma 3.2, $\Sigma_S = \Sigma_T = \operatorname{diag}(a_n, a_n, b_n)$ with $a_n \neq b_n$, so ψ commutes with Σ_S and preserves its eigenspaces, the plane $z = 0$ and the z-axis: $\psi = \operatorname{diag}(B, \varepsilon)$ with $B \in O(2)$ and $\varepsilon = \pm 1$. The projection of both point sets to the plane is the grid, so B is a symmetry of the grid, that is, an element $g \in D_4$; hence $\psi = \hat{g}$ or $\psi = \hat{g}\tau$. By (2), $\hat{g}(P_S) = P_{gS}$ and $\hat{g}\tau(P_S) = P_{g\overline{S}}$, and

the point set determines the arrangement; this proves (1), and (2) follows. For (3) take $T = S$: $\hat{g}(P_S) = P_S$ forces $gS = S$, hence $g = \mathrm{id}$, and $\hat{g}\tau(P_S) = P_S$ forces $g\overline{S} = S$, i.e. $\overline{S} = g^{-1}S = gS$ by Lemma 4.1. □

So for every order the symmetries of the point set of a semi-magic square are exactly those coming from self-complementarity, and congruences between point sets come only from D_4 and complementation.

4.2 The isometry groups of the order-four gems

For the gem, $\mathrm{Iso}(\mathcal{G}(S)) = \mathrm{Iso}(\mathrm{vert}\,\mathcal{G}(S))$, since isometries preserve extreme points, and $\mathrm{Iso}(P_S) \subseteq \mathrm{Iso}(\mathcal{G}(S))$; the inclusion is an equality when every point is a vertex, but in general the gem might have isometries that do not preserve the non-vertex points. For order four this does not happen.

Theorem 4.3 (Isometry groups of the order-four gems; exact computation)**.** *For each of the* 880 *D_4-classes of magic squares S of order* 4, $\mathrm{Iso}(\mathcal{G}(S)) = \mathrm{Iso}(P_S)$. *Hence:*

1. *for the* 48 *associative squares,* $\mathrm{Iso}(\mathcal{G}(S)) = \{\mathrm{id}, -\mathrm{id}\}$*;*
2. *for the* 304 *self-complementary squares that are not associative,* $\mathrm{Iso}(\mathcal{G}(S)) = \{\mathrm{id}, \iota_g\}$ *with* $g \in \{h, v\}$*, a half-turn about a horizontal coordinate axis;*
3. *for the remaining* 528 *squares,* $\mathrm{Iso}(\mathcal{G}(S))$ *is trivial.*

In particular a gem has an improper isometry if and only if S is associative, and the 832 *gems in cases (2) and (3) are chiral (chiral: having no orientation-reversing isometry).*

Proof. The squares were sorted into the three cases by comparing $\overline{S}$ with the eight images gS; by Lemma 4.1 the complementing element is r^2, h or v. For each square we computed $\mathrm{Iso}(\mathrm{vert}\,\mathcal{G}(S))$ in exact integer arithmetic (Appendix A); its order is 2 in cases (1) and (2) and 1 in case (3), equal to the order of $\mathrm{Iso}(P_S)$ by Theorem 4.2(3), and $\mathrm{Iso}(P_S) \subseteq \mathrm{Iso}(\mathcal{G}(S))$. The maps ι_h, ι_v are proper. □

The partition (1)–(3) is classical: the associative squares form Dudeney's group III and the self-complementary non-associative ones his group VI [11]. What Theorem 4.3 adds is that the hulls have no further symmetry. Likewise the congruence classes of the 880 gems are the orbits of complementation (Proposition 5.4). Table 1 summarises the three cases and Figure 6 shows one gem of each.

Table 1: Symmetry types of the 880 order-four gems; "classes" counts congruence classes.

type of S	Dudeney	squares	$\mathrm{Iso}(\mathcal{G})$	classes	V
associative	III	48	$\{\pm\mathrm{id}\}$	48	$10, 12, 14, 16$
self-complementary, not associative	VI	304	$\{\mathrm{id}, \iota_g\}$, half-turn	304	$8, 10, 12, 14, 16$
not self-complementary	others	528	trivial	264	$8, \ldots, 16$
total		880		616	

Lemma 4.4 (Parity for associative squares)**.** *Let S be associative of order n and not a D_4-image of the natural square (for instance, associative and semi-magic). Then V and V^∂ are even and* $V^\circ \equiv n \pmod 2$.

Proof. The point set is invariant under $-\mathrm{id}$ (Lemma 4.1), which therefore permutes the vertices, the interior points and the non-vertex boundary points of the 3-polytope $\mathcal{G}(S)$ (Proposition 2.3). Its only fixed point is $\mathbf{0}$, which is a point $\mathbf{p}_c$ exactly when n is odd (the centre cell), and then an interior point. □

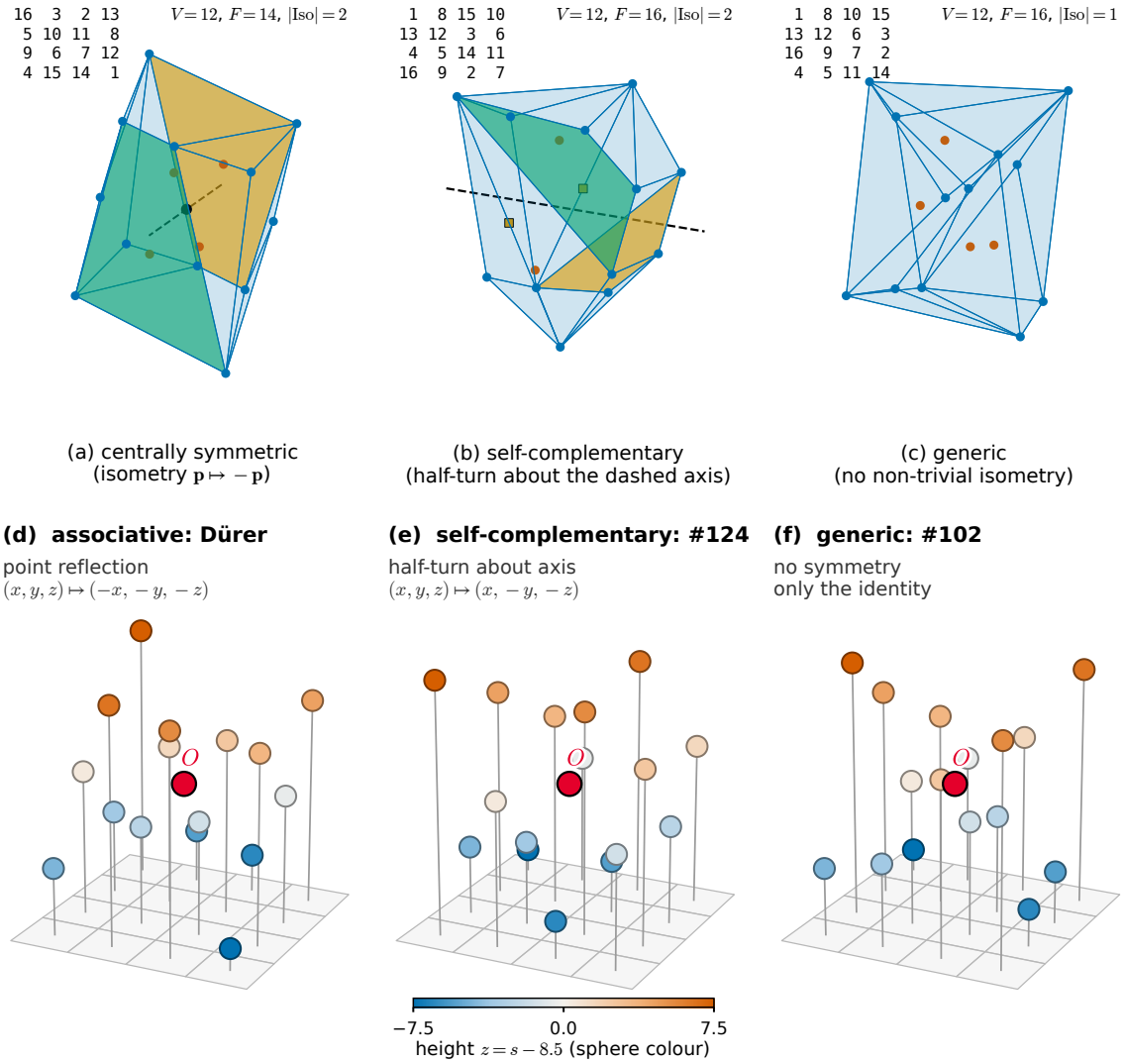


Figure 6: Gems (top) and point sets (bottom) of three order-four squares, all drawn from the same viewpoint, with the square printed above each gem. Interior points: red discs; non-vertex boundary points: orange squares. (a, d) Dürer's square, associative: a pair of opposite facets (orange, green), and the 8 antipodal pairs $\{\mathbf{p}, -\mathbf{p}\}$ joined through O. (b, e) Square #124, self-complementary and not associative: the half-turn about the dashed axis exchanges the highlighted facets and the 8 pairs of points. (c, f) Square #102: no non-trivial isometry; the gem is chiral and its mirror image in $z = 0$ is the gem of $17 - S$.

The hypothesis is needed: the natural square of order 3 is associative, but its gem is a parallelogram and has no interior points. For $n = 4$ the 48 associative gems have $V^\circ \in \{0, 2, 4\}$ (4, 23 and 21 gems); among the other squares both parities occur.

Remark 4.5 (Facet orbits). For an associative square, $-\mathrm{id}$ fixes no facet, since no facet plane passes through the interior point $\mathbf{0}$; so F is even. For the half-turn ι_h (the case ι_v is the same after the reflection d), ι_h maps the plane $\mathbf{a} \cdot \mathbf{p} = b$ to the plane $\iota_h(\mathbf{a}) \cdot \mathbf{p} = b$. The two coincide if and only if $\iota_h(\mathbf{a}) = \mathbf{a}$ ($\mathbf{a}$ parallel to the axis), or $\iota_h(\mathbf{a}) = -\mathbf{a}$ and $b = 0$. The second case is a plane through $\mathbf{0}$, which cannot support the gem, since $\mathbf{0}$ is interior (Proposition 2.3). So an ι_h-invariant facet plane is one of the vertical planes $y = \pm\frac{n-1}{2}$ over the top and bottom rows; by Proposition 5.2(2) below, these carry facets unless the row is an arithmetic progression. Among the 304 order-four gems of case (2), 282 have two fixed facets, 18 one and 4 none. Neither isometry fixes a vertex, so V is even for all 352 self-complementary squares.

Remark 4.6 (Difference-body ratio). For a convex body $K \subset \mathbb{R}^3$ with non-empty interior let $\sigma(K) = \mathrm{vol}(K - K)/(8 \,\mathrm{vol}\, K)$, the *difference-body ratio*. By the Brunn–Minkowski inequality [30], $\sigma(K) \geq 1$, with equality if and only if K is centrally symmetric; by the Rogers–Shephard

inequality [29], $\sigma(K) \le \binom{6}{3}/8 = \frac{5}{2}$, with equality if and only if K is a simplex; see [16, 30] for this and other measures of symmetry. By Theorem 4.3, $\sigma = 1$ exactly for the 48 associative gems of order four; for the other 832, an exact computation (`anc/difference_body_ratio.py`) gives $\sigma \in [1057/1018, 145/88] \approx [1.038, 1.648]$.

5 The exact geometry of the order-four gems

The isometry groups of Section 4 are one invariant of a gem. This section computes the others, exactly, for all 880 gems of order four: vertices, faces, volumes, combinatorial types and congruence classes. To read these numbers one needs two facts that hold for every order: some points are vertices of every gem, and the faces of a gem are two regular subdivisions of the grid.

5.1 Forced vertices and regular subdivisions

The *sides* of the grid are its first and last rows and columns. Every gem lies in the box $\mathcal{B}_n = [-\frac{n-1}{2}, \frac{n-1}{2}]^2 \times [-\frac{N-1}{2}, \frac{N-1}{2}]$ and meets each of its six faces: the top and the bottom face in the points of the entries N and 1, and the vertical face over a side e of the grid in the polygon $\operatorname{conv}\{\mathbf{p}_c : c \in e\}$. Each contact is a face of $\mathcal{G}(S)$, so its vertices are vertices of $\mathcal{G}(S)$; the lemma below names those that can be read off from the entries.

Lemma 5.1 (Forced vertices)**.** *Let S be an arrangement of order $n \ge 2$. The following points are vertices of $\mathcal{G}(S)$; we call them the* forced vertices *of S: the points over the four corner cells; the points carrying the entries 1 and N; and, for each side, the points over the cells of the smallest and of the largest entry on that side.*

Proof. A point that is the unique maximiser of a linear functional over the N points is a vertex. The entries are distinct, so $\mp z$ have unique maximisers where $s = 1$ and $s = N$. The functional $-x + y$ attains its maximum only at the top-left corner, and similarly for the other corners. On the left column $x = -\frac{n-1}{2}$, while $x \ge -\frac{n-1}{2} + 1$ elsewhere; for $\varepsilon = 1/N$ the functional $-x + \varepsilon z$ exceeds $\frac{n-1}{2} - \frac{1}{2}$ on the left column and is less than $\frac{n-1}{2} - \frac{1}{2}$ elsewhere, so its unique maximiser is the point of largest height on the left column; $-x - \varepsilon z$ gives the smallest, and the other sides follow by symmetry. □

For order four the forced vertices number between 4 and 13, and in general other points are vertices as well (Section 5.2).

Gems and regular subdivisions. The gem is two regular subdivisions of the grid, by z and by $-z$, glued along the boundary of the grid (with vertical facets over the sides where the entries are not in arithmetic progression); Figure 3e shows the two subdivisions for the Lo Shu. Let A be the $n \times n$ grid of cell positions. For a height function $\omega\colon A \to \mathbb{R}$, the *regular subdivision* $\mathcal{S}(\omega)$ of A is obtained by projecting to the plane the lower faces of $\operatorname{conv}\{(c, \omega(c))\}$, i.e. the faces whose outer normals have negative last coordinate; its cells are the sets of grid points whose lifts lie on a common lower face [9, Ch. 2], [15, Ch. 7]. Cells are marked: a cell is a set of grid points, and its convex hull may contain grid points that are not in it. $\mathcal{S}(\omega)$ does not change if ω is multiplied by a positive scalar or an affine function of (x, y) is added to it; the height functions inducing a given subdivision form a relatively open polyhedral cone, its *secondary cone*. By definition, $\mathcal{G}(S)$ is the lift of A by the height function z of S, and its upper faces are the reflections in $z = 0$ of the lower faces of the lift by $-z$.

Proposition 5.2 (Gems as pairs of regular subdivisions)**.** *Let S be an arrangement of order $n \ge 2$ that is not a D_4-image of the natural square.*

1. *Every facet of $\mathcal{G}(S)$ is a lower facet, an upper facet or a vertical facet. Projection to the plane maps the lower facets onto the maximal cells of $\mathcal{S}(z)$ and the upper facets onto the maximal cells of $\mathcal{S}(-z)$.*

2. *The vertical facets lie in the four planes* $x = \pm\frac{n-1}{2}$, $y = \pm\frac{n-1}{2}$; *the plane over a side of the grid contains a facet if and only if the entries along that side, in cell order, are not an arithmetic progression.*

3. $\mathbf{p}_c$ *is a vertex of* $\mathcal{G}(S)$ *if and only if* c *is a vertex of a cell of* $\mathcal{S}(z)$ *or of* $\mathcal{S}(-z)$.

4. *The face lattice of* $\mathcal{G}(S)$, *with vertices labelled by cells, is determined by the pair* $(\mathcal{S}(z), \mathcal{S}(-z))$, *hence by the secondary cones containing* z *and* $-z$. *The complement* $N + 1 - S$ *exchanges the two subdivisions.*

Proof. (1) is the definition of $\mathcal{S}(z)$ and, after reflecting in $z = 0$, of $\mathcal{S}(-z)$. (2) A vertical facet lies in a plane $\ell(x, y) = \max_A \ell$ for a linear form ℓ; the maximisers of ℓ on the grid are a corner or the n cells of a side, and the face is 2-dimensional only in the second case and only if the lifts of the side are not collinear, i.e. if the heights along the side are not an affine function of the position. (3) The vertical line through a vertex $\mathbf{p}_c$ meets $\mathcal{G}(S)$ in a segment with endpoint $\mathbf{p}_c$, so $\mathbf{p}_c$ lies on a lower or an upper face, of which it is a vertex; conversely the vertices of a face are vertices of the polytope. (4) Let U and L be the upper and lower boundary functions of $\mathcal{G}(S)$ over $\operatorname{conv} A$. $U - L$ is concave, non-negative, and positive at the projection of an interior point, hence positive on $\operatorname{int}\operatorname{conv} A$. So a lower face and an upper face meet only over $\partial \operatorname{conv} A$. Over a side e of the grid, $\mathcal{G}(S)$ meets the vertical plane in $\operatorname{conv}\{\mathbf{p}_c : c \in e\}$. This polygon is bounded by the restrictions of the two boundary chains to e, and it is a vertical facet exactly when these chains differ, i.e. (by (2)) when the entries along e are not an arithmetic progression. A lower or upper facet projects bijectively onto its cell, so its vertices are the lifts of the vertices of the convex hull of the cell. Hence the marked cells of $\mathcal{S}(z)$ and $\mathcal{S}(-z)$, together with the vertical facets, determine every vertex–facet incidence. The complement has height function $-z$. □

Example 5.3 (Lo Shu, continuing Example 2.4)**.** Writing each grid point as its entry, $\mathcal{S}(z)$ has the maximal cells $\{1, 2, 6\}$, $\{1, 3, 8\}$, $\{1, 2, 3, 4\}$, and $\mathcal{S}(-z)$ has $\{2, 7, 9\}$, $\{4, 8, 9\}$, $\{6, 7, 8, 9\}$. The vertical facets are the triangles $\{2, 4, 9\}$, $\{2, 6, 7\}$, $\{1, 6, 8\}$, $\{3, 4, 8\}$. The centre point (height 0) lies in the convex hull of each quadrilateral cell but belongs to no cell: the plane of $\{1, 2, 3, 4\}$ is $z = -x - y - 3$, which is $-3 < 0$ at the centre.

In this language, the four corner cells are vertices of $\operatorname{conv} A$ and so give vertices of every gem. The other forced vertices of Lemma 5.1 are, for each side e of the grid, the lowest and the highest lifted point over e; these are vertices of the polygon $\operatorname{conv}\{\mathbf{p}_c : c \in e\}$, the face of $\mathcal{G}(S)$ over e. For the 880 order-four gems, 856 have four vertical facets, 20 have three and 4 have two; the height functions of the 7040 order-four magic squares induce 417 distinct regular subdivisions of the 4×4 grid up to D_4, and for 548 of the 880 classes $\mathcal{S}(z)$ is a triangulation (`anc/verify_structure.py`). The combinatorial types of Section 5.2 are a coarse invariant of the pair $(\mathcal{S}(z), \mathcal{S}(-z))$ (face lattices without cell labels). In the proof of the Siamese theorem (Section 6), the shear Φ of Step 3 uses the fact that $\mathcal{S}(\pm z)$ is unchanged when an affine function of (x, y) is subtracted from z and the result is divided by n.

5.2 The census

There are 8 magic squares of order 3, all images of the Lo Shu under D_4, and 7,040 of order 4, forming 880 D_4-classes; the order-four squares were listed by Frénicle de Bessy [13] and have been recounted by computer [5]. Dudeney sorted them into twelve groups according to the pattern of complementary pairs $(a, 17 - a)$ [11]; since the numbering of the groups differs between reproductions of his table, we list the labels used here, for all 7040 squares together with the number # of their class, in `anc/all7040_dudeney.tsv`. The number 275,305,224 of magic squares of order five up to D_4 is due to Schroeppel (1973) [14]; for order six the count was estimated by Monte Carlo methods [26] and was computed exactly by H. Mino in 2024, as announced on his web page [21] and reported in [33].

For each of the 880 squares we computed $\mathcal{G}(S)$ exactly (integer facet inequalities, rational volumes; Appendix A). Table 2 and Figure 7 summarise the results.

Table 2: The 880 order-four gems by number V of vertices: numbers of squares, of combinatorial types (face lattices up to isomorphism), of f-vectors and of distinct exact volumes, and the volume as a fraction of the box $\mathcal{B}_4$ ($\operatorname{vol}\mathcal{B}_4 = 135$). A volume can occur for several values of V, so the volume counts add up to more than 155. Bottom: numbers of gems with k interior points ($V^\circ = k$) and with k non-vertex boundary points ($V^\partial = k$).

V	8	9	10	11	12	13	14	15	16	all
squares	7	10	103	80	380	106	162	14	18	880
types	4	1	27	32	129	52	85	7	15	352
f-vectors	4	1	6	5	7	5	5	3	3	39
volumes	4	5	51	33	111	40	65	6	14	155
min of $\operatorname{vol}/\operatorname{vol}\mathcal{B}_4$	.622	.588	.481	.549	.430	.456	.452	.498	.449	.430
mean of $\operatorname{vol}/\operatorname{vol}\mathcal{B}_4$	.642	.623	.631	.630	.634	.614	.608	.560	.536	.623
max of $\operatorname{vol}/\operatorname{vol}\mathcal{B}_4$	.648	.667	.701	.699	.721	.688	.691	.615	.598	.721

k	0	1	2	3	4	5	6
gems with $V^\circ = k$	37	32	353	178	280	0	0
gems with $V^\partial = k$	357	204	257	40	21	0	1

Vertices, faces and volumes. Between 8 and all 16 points are vertices; $V = 12$ is the most frequent value (380 gems), and the mean is $269/22 \approx 12.23$. The 39 f-vectors range from $(8, 14, 8)$ (one gem) to $(16, 36, 22)$ (nine gems); the most frequent is $(12, 26, 16)$ (164 gems). The volume takes 155 distinct values, from 58 to $292/3$, with mean 84.05; its denominators are $1, 2, 3, 6$, because for even n all coordinates are half-odd integers, so that $\mathcal{G}(S) + (\frac{1}{2}, \frac{1}{2}, \frac{1}{2})$ is a lattice polytope. The Ehrhart polynomials of these lattice polytopes take 554 distinct values (Appendix B). The gems of order four fill between 43% and 72% of the box $\mathcal{B}_4$ (Section 5.1). At most four points are interior, and at most six are non-vertex boundary points.

Combinatorial types. Two gems have the same combinatorial type if their vertex–facet incidence graphs are isomorphic by an isomorphism that maps vertices to vertices and facets to facets. Testing pairs of incidence graphs for isomorphism gives 352 types (Appendix A). By (4) the gems of S and $\overline{S}$ are mirror images, so every type is a union of complement orbits; the 93 types with one member are self-complementary squares, and of the 201 types with two members, 177 are mirror pairs and 24 consist of two self-complementary squares. The numbers of types with $1, 2, 3, \ldots$ members are

squares per type	1	2	3	4	5	6	7	8	10	13	14	16	26	32
types	93	201	8	20	4	7	4	5	5	1	1	1	1	1

The largest type, with $f = (10, 22, 14)$ (twelve triangles and two quadrilaterals), has 32 members. Each of Dudeney's groups with more than eight members contains gems with at least three different vertex counts and at least 20 combinatorial types, and the 8-vertex gems occur only in groups VI (three) and VIII (four). The 48 pandiagonal squares (Dudeney's group I [24]) have gems with 11 to 15 vertices and between one and four interior points; no order-four square is both pandiagonal and associative.

Proposition 5.4 (The vertex set determines the square; congruence)**.** *Let S and T be magic squares of order four.*

1. *If $\hat{g}\operatorname{vert}\mathcal{G}(S) = \operatorname{vert}\mathcal{G}(T)$ for some $g \in D_4$, then $T = gS$. In other words, the vertex set of the gem, up to the action of D_4, determines the D_4-class of the square.*

2. *$\mathcal{G}(S)$ and $\mathcal{G}(T)$ are congruent if and only if $T \in D_4S \cup D_4\overline{S}$. There are* 616 *congruence classes of gems:* 352 *self-complementary classes and* 264 *mirror pairs.*

Proof. (1) The 880 vertex sets have pairwise distinct D_4-canonical forms (exact comparison), so $T = g'S$ for some $g' \in D_4$, and then $\hat{g}'\operatorname{vert}\mathcal{G}(S) = \operatorname{vert}\mathcal{G}(T) = \hat{g}\operatorname{vert}\mathcal{G}(S)$ by (2). Thus $\hat{g}^{-1}\hat{g}'$ is an isometry of $\mathcal{G}(S)$ that preserves heights; by Theorem 4.3 the only such isometry is the identity, so $g' = g$. (2) "If" is (2) and (4). For "only if": the sorted multiset of pairwise squared distances of the vertex set is a congruence invariant, and it takes 616 distinct values over the 880 gems (exact computation). Gems in one complement orbit are congruent, and there are 616 orbits (352 self-complementary classes and 264 pairs). Hence the congruence classes are exactly the complement orbits. □

Forced vertices and named squares. For order four nothing beyond the forced vertices of Lemma 5.1 is a vertex of every gem: the only cells that carry a vertex in all 880 gems are the four corners, and the only entries that always label a vertex are 1 and 16. In eight gems the point of the entry 2 is a non-vertex boundary point, for instance for square #593 below. The gems of the Lo Shu, of Dürer's square and of the Siamese square of order 5 have 8 of 9, 12 of 16 and 12 of 25 points as vertices and volumes 18, 92 and 830/3. The gem of Dürer's square is centrally symmetric with twelve vertices; exchanging its first two rows and its last two rows gives square #802, whose gem has all sixteen points as vertices. The squares cited in Figures 5, 6 and 7 and in the text are

$$\#45 = \begin{pmatrix} 1 & 6 & 11 & 16 \\ 8 & 15 & 2 & 9 \\ 12 & 3 & 14 & 5 \\ 13 & 10 & 7 & 4 \end{pmatrix} \quad \#54 = \begin{pmatrix} 1 & 6 & 11 & 16 \\ 15 & 12 & 5 & 2 \\ 14 & 9 & 8 & 3 \\ 4 & 7 & 10 & 13 \end{pmatrix} \quad \#102 = \begin{pmatrix} 1 & 8 & 10 & 15 \\ 13 & 12 & 6 & 3 \\ 16 & 9 & 7 & 2 \\ 4 & 5 & 11 & 14 \end{pmatrix} \quad \#124 = \begin{pmatrix} 1 & 8 & 15 & 10 \\ 13 & 12 & 3 & 6 \\ 4 & 5 & 14 & 11 \\ 16 & 9 & 2 & 7 \end{pmatrix}$$

$$\#379 = \begin{pmatrix} 2 & 12 & 5 & 15 \\ 13 & 7 & 10 & 4 \\ 16 & 6 & 11 & 1 \\ 3 & 9 & 8 & 14 \end{pmatrix} \quad \#593 = \begin{pmatrix} 4 & 1 & 16 & 13 \\ 5 & 15 & 2 & 12 \\ 14 & 8 & 9 & 3 \\ 11 & 10 & 7 & 6 \end{pmatrix} \quad \#802 = \begin{pmatrix} 5 & 10 & 11 & 8 \\ 16 & 3 & 2 & 13 \\ 4 & 15 & 14 & 1 \\ 9 & 6 & 7 & 12 \end{pmatrix}$$

6 Larger orders: Siamese gems

Beyond order four an exhaustive computation is out of reach: there are 275,305,224 magic squares of order five up to D_4 (Section 5.2). Samples suggest that the number of vertices grows with n, for random arrangements and for magic squares alike (Table 4, numerical). The classical construction of de la Loubère behaves in the opposite way: for the Siamese square of every odd order $n \geq 5$ the vertices of the gem are exactly the twelve forced vertices of Lemma 5.1 (Theorem 6.2). The proof combines the tools of the previous sections: an explicit closed form for the square, the invariance of the subdivisions $\mathcal{S}(\pm z)$ under adding affine functions (Section 5.1), the forced vertices, and, for the isometry group, Theorem 4.2.

6.1 Two linear magic squares

The *Siamese* (de la Loubère) square of odd order $n = 2m+1$ [8] has 1 in the middle cell of the top row; each next entry is placed one step up and one step to the right, cyclically, or one step down if that cell is occupied. For $n = 5$ it is

$$\begin{pmatrix} 17 & 24 & 1 & 8 & 15 \\ 23 & 5 & 7 & 14 & 16 \\ 4 & 6 & 13 & 20 & 22 \\ 10 & 12 & 19 & 21 & 3 \\ 11 & 18 & 25 & 2 & 9 \end{pmatrix},$$

and for $n = 3$ it is a D_4-image of the Lo Shu. Write $\langle t \rangle \in \{0, \ldots, n-1\}$ for the residue of t modulo n.

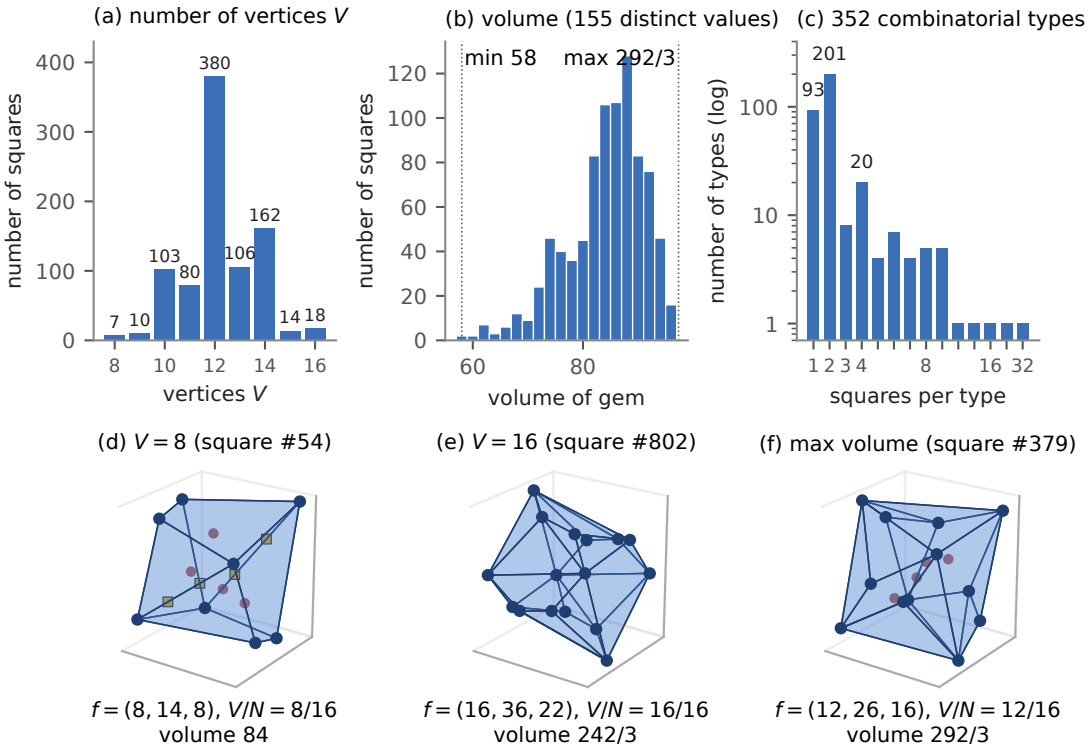


Figure 7: The 880 order-four gems. (a) Number of vertices. (b) Exact volume. (c) Sizes of the 352 combinatorial types (logarithmic scale). (d)–(f) Three extreme gems with translucent faces: the unique gem with $f = (8, 14, 8)$ (square #54; interior points as red discs, non-vertex boundary points as orange squares), a gem with all 16 points as vertices (square #802), and the gem of largest volume (square #379).

Lemma 6.1 (Two linear magic squares)**.** *For odd $n = 2m + 1$, both*

$$s(i, j) = n\langle i + j - m\rangle + \langle i + 2j - 2m\rangle + 1 \quad \text{and} \quad s'(i, j) = n\langle i + 2j - 2m\rangle + \langle i + j - m\rangle + 1$$

are magic squares. We call s the closed form of the Siamese square (Step 1 of the proof of Theorem 6.2) and s' the digit-exchanged square*.*

Proof. The map $(i, j) \mapsto (\langle i + j - m\rangle, \langle i + 2j - 2m\rangle)$ is a bijection of $\mathbb{Z}_n^2$ (its linear part has determinant 1), so both squares are bijections onto $\{1, \dots, n^2\}$. In every row and every column both digits run over $\mathbb{Z}_n$, since 1 and 2 are units modulo n. On the anti-diagonal $j = 2m - i$ one digit is the constant $\langle m\rangle = m$ and the other is $\langle 2m - i\rangle$, a permutation; n copies of m sum to $n(n - 1)/2$, as a permutation does. On the main diagonal the digits are $\langle 2i - m\rangle$, a permutation since n is odd, and $\langle 3i - 2m\rangle$. The latter is a permutation if $3 \nmid n$. If $3 \mid n$ then $m \equiv 1 \pmod 3$, so $\langle 3i - 2m\rangle$ runs three times over $\{1, 4, \dots, n - 2\}$, whose sum is again $n(n - 1)/2$. Hence every line sum is $n \cdot n(n - 1)/2 + n(n - 1)/2 + n = \mu_n$. □

6.2 The theorem

Theorem 6.2 (Siamese gems)**.** *Let S be the Siamese square of odd order $n = 2m + 1 \geq 3$.*

(a) For $n = 3$ (the Lo Shu up to D_4) the vertices of $\mathcal{G}(S)$ are the eight boundary points, $f = (8, 16, 10)$ (Example 2.4), and $\mathrm{Iso}(\mathcal{G}(S)) = \{\pm\mathrm{id}\}$*.*

(b) For $n \geq 5$ the vertices of $\mathcal{G}(S)$ are exactly the twelve forced vertices of Lemma 5.1, the points over the cells

$$\begin{array}{lll} (0,0)\colon \bar{s} + m + 2, & (0, m-1)\colon N - 1, & (0, m)\colon 1, \\ (0, 2m)\colon \bar{s} + m, & (m-1, 0)\colon N - m, & (m, 0)\colon m + 2, \\ (m, 2m)\colon N - m - 1, & (m+1, 2m)\colon m + 1, & (2m, 0)\colon \bar{s} - m, \\ (2m, m)\colon N, & (2m, m+1)\colon 2, & (2m, 2m)\colon \bar{s} - m - 2, \end{array}$$

listed with their entries; $f = (12, 26, 16)$*, with* 12 *triangular and* 4 *quadrilateral facets; and* $\mathrm{Iso}(\mathcal{G}(S)) = \mathrm{Iso}(P_S) = \{\pm\mathrm{id}\}$.

So only 12 of the n^2 points are vertices, and the combinatorial type of the gem does not depend on $n \geq 5$.

Proof. Step 1: closed form. We show that S is the square $s(i,j) = n\langle i+j-m\rangle + \langle i+2j-2m\rangle + 1$ of Lemma 6.1; this identification with de la Loubère's construction is classical [2]. The map $\kappa(i,j) = (\langle i+j-m\rangle, \langle i+2j-2m\rangle)$ is a bijection of $\mathbb{Z}_n^2$, and $\kappa(0,m) = (0,0)$. Write $k-1 = nk_1+k_0$ with $0 \leq k_0 < n$; we prove by induction on k that the entry k is placed in the cell $\kappa^{-1}(k_1,k_0)$. This holds for $k=1$. Suppose it holds for $1,\dots,k$, with k in the cell $c = \kappa^{-1}(k_1,k_0)$; the occupied cells are $\kappa^{-1}(k_1',k_0')$ with $nk_1'+k_0' < k$. The up-right move $(i,j) \to (i-1,j+1)$ keeps $i+j$ and increases $i+2j$ by 1, so it leads to $\kappa^{-1}(k_1,\langle k_0+1\rangle)$. If $k_0 < n-1$, this cell is empty and is the cell prescribed for $k+1$. If $k_0 = n-1$, i.e. $n \mid k$, it is the cell $\kappa^{-1}(k_1,0)$ of the entry nk_1+1, so the move is blocked; the downward move $(i,j) \to (i+1,j)$ increases both $i+j$ and $i+2j$ by 1 and leads to the empty cell $\kappa^{-1}(k_1+1,0)$, the cell prescribed for $k+1 = n(k_1+1)+1$. (Thus each run of n entries fills one broken diagonal $i+j \equiv \mathrm{const}$.) By Lemma 6.1, S is magic. It is also associative: by the closed form, $s(i,j)+s(2m-i,2m-j) = N+1$.

Step 2: the forced vertices. In every row and every column the leading digit $\langle i+j-m\rangle$ takes each value in $\{0,\dots,n-1\}$ once, so the largest entry of a side is in the cell where this digit is $n-1$ and the smallest where it is 0. On the top row $i=0$ this gives $j = m-1$ with entry $n(n-1)+\langle -2\rangle+1 = N-1$, and $j=m$ with entry 1; on the bottom row $i=2m$, $j=m$ with entry $n(n-1)+n = N$, and $j=m+1$ with entry $\langle n+1\rangle+1 = 2$; on the left column $j=0$, $i=m-1$ with entry $n(n-1)+\langle -m-1\rangle+1 = N-m$, and $i=m$ with $\langle -m\rangle+1 = m+2$; on the right column $j=2m$, $i=m$ with entry $n(n-1)+\langle 3m\rangle+1 = N-m-1$, and $i=m+1$ with $\langle 3m+1\rangle+1 = m+1$. The corner entries follow in the same way. The entries 1 and N are among the side extremes, and for $m \geq 2$ the eight side cells are distinct and are not corners; so the forced vertices are the twelve listed points. For $m=1$ the side extremes are the corners and the midpoints of the sides.

Proof of (a). For $n=3$ the forced vertices are the eight boundary points, and the only other point is the interior point $\mathbf{0}$ (Example 2.4). An isometry of $\mathcal{G}(S)$ maps its vertex set onto itself and so fixes the centroid $\mathbf{0}$ of the vertex set; hence it preserves P_S, and $\mathrm{Iso}(P_S) = \{\pm\mathrm{id}\}$ by Theorem 4.2, since S is associative. From now on let $m \geq 2$. It remains to show that no point other than the twelve forced vertices is a vertex.

Step 3: shear to five levels. With $x = j-m$, $y = m-i$ and $\langle t\rangle = t - n\lfloor t/n\rfloor$ the closed form gives

$$z = \lambda(x,y) + n\,\zeta(x,y), \qquad \lambda(x,y) = n(x-y+m) + (2x-y+m) + 1 - \bar{s},$$
$$\zeta(x,y) = -n\left\lfloor \tfrac{\ell_1}{n}\right\rfloor - \left\lfloor \tfrac{\ell_2}{n}\right\rfloor, \qquad \ell_1 = x-y+m, \quad \ell_2 = 2x-y+m = \ell_1 + x.$$

The map $\Phi(x,y,z) = (x,y,(z-\lambda(x,y))/n)$ is a shear (an affine bijection that preserves vertical lines), so $\Phi(\mathcal{G}(S)) = \mathrm{conv}\{(x,y,\zeta(x,y))\}$ has the same face lattice and vertex cells as $\mathcal{G}(S)$; equivalently, $\mathcal{S}(\pm z) = \mathcal{S}(\pm\zeta)$. After the shear the height ζ takes five values, and its level sets are the lattice points of five polygons Λ_k (the *level polygons*). Indeed, for $|x|,|y| \leq m$ we have $-m \leq \ell_1 \leq 3m$ and $-2m \leq \ell_2 \leq 4m$, so both floors lie in $\{-1,0,1\}$. If $\ell_1 < 0$, then $x \leq y-m-1 \leq -1$, so $\ell_2 < 0$ and $\zeta = n+1$. If $\ell_1 \geq n$, then $x \geq y+m+1 \geq 1$, so $\ell_2 \geq n$ and $\zeta = -(n+1)$. If $0 \leq \ell_1 \leq 2m$, then $\lfloor \ell_2/n\rfloor = -1,0,1$ according as $y \geq 2x+m+1$, $|y-2x| \leq m$, or $y \leq 2x-m-1$. Conversely, if $|y-2x| \leq m$, then $\ell_1 < 0$ is impossible, since it would force $\ell_2 < 0$, i.e. $y \geq 2x+m+1$; likewise $\ell_1 \geq n$ would force $y \leq 2x-m-1$. So, in the box $|x|,|y| \leq m$ (Figure 8a): $\zeta = n+1$ on $\Lambda_{+2} = \{y \geq x+m+1\} = \mathrm{conv}\{(-m,m),(-1,m),(-m,1)\}$; $\zeta = 1$ on $\Lambda_{+1} = \{2x+m+1 \leq y \leq x+m\} = \mathrm{conv}\{(-m,0),(-m,1-m),(-1,m-1)\}$; $\zeta = 0$ on $\Lambda_0 = \{|y-2x| \leq m\} = \mathrm{conv}\{(-m,-m),(0,m),(m,m),(0,-m)\}$; and $\zeta = -1$, $-(n+1)$ on $\Lambda_{-1} = -\Lambda_{+1}$, $\Lambda_{-2} = -\Lambda_{+2}$.

Step 4: the hull. Let $q_1, \dots, q_{12}$ be the images under Φ of the twelve forced vertices and $q_{13}, \dots, q_{16}$ the remaining four vertices of the level polygons; their coordinates are in Table 3. Let H be the polyhedron defined by the eight inequalities below and their images (1′)–(8′) under $-\mathrm{id}$. Inequality (k) is the plane through three of the twelve points, namely (1) $q_1q_2q_3$, (2) $q_1q_2q_8$, (3) $q_1q_3q_5$, (4) $q_1q_4q_7$, (5) $q_2q_3q_6$, (6) $q_2q_6q_9$, (7) $q_2q_8q_9$, (8) $q_3q_5q_6$; its normal is the cross product of two edge vectors:

$$\begin{aligned}
&(1)\ \zeta \le 2m+2, \qquad (2)\ y \le m, \qquad (3)\ -x \le m, \qquad (8)\ -2y+\zeta \le 2m,\\
&(4)\ -2(m{+}1)x+(2m{+}1)y-m\zeta \le m(2m{+}1), \qquad (7)\ 2mx-y+m\zeta \le m(2m{-}1),\\
&(5)\ 2(m{+}1)x-2(m{+}1)y+(2m{+}1)\zeta \le 2m(m{+}1),\\
&(6)\ 2m(m{+}2)x-(2m^2{+}2m{-}1)y+m(2m{+}1)\zeta \le m(2m^2{+}2m{-}1).
\end{aligned}$$

Table 3 lists the slack (right-hand side minus left-hand side) of (1)–(8) at the sixteen points q_k. Every non-zero entry is a product of positive constants, of linear factors $(m-1,\ m,\ m+1,\ m+2,\ 2m-1,\ 2m+1,\ 2m+3,\ 4m-3,\ 4m-1)$, which are positive for $m \ge 2$, and of the quadratics $2m^2+2m+1$, $2m^2+2m-1$, $4m^2+2m-1$, $4m^2-2m+1$, $4m^2+m+1$, $4m^2-3m+1$, m^2+m+1. Each of these quadratics is positive at $m=1$ and has its minimum at some $m \le 3/8$, hence is positive for all $m \ge 1$. The set $\{q_1, \dots, q_{16}\}$ is centrally symmetric, so all sixteen inequalities hold at all sixteen points and hence, by convexity, at all points (x, y, ζ), each of which lies in a level polygon: $\Phi(\mathcal{G}(S)) \subseteq H$. The zeros of the table show that the planes (1)–(8) contain the following q_k $(k \le 12)$, in cyclic order:

$$\begin{aligned}
&(1)\colon q_1q_2q_3, \quad (2)\colon q_1q_2q_8q_7, \quad (3)\colon q_1q_3q_5q_4, \quad (4)\colon q_1q_4q_7,\\
&(5)\colon q_2q_3q_6, \quad (6)\colon q_2q_6q_9, \quad (7)\colon q_2q_8q_9, \quad (8)\colon q_3q_5q_6,
\end{aligned}$$

with the antipodal lists for (1′)–(8′) ($q_{10}=-q_1$, $q_{11}=-q_3$, $q_{12}=-q_2$, $q_9=-q_4$, $q_8=-q_5$, $q_7=-q_6$). All four points of each quadrilateral are forced vertices, hence in convex position, so only the cyclic order needs checking. In the plane $y=m$, written in coordinates (x,ζ), $q_1=(-m,2m+2)$, $q_2=(-1,2m+2)$, $q_8=(m,0)$ and $q_7=(0,0)$ form a trapezoid whose parallel sides q_1q_2 and q_8q_7 are traversed in opposite directions. In the plane $x=-m$, written in coordinates (y,ζ), the line through $q_5=(-m,0)$ and $q_1=(m,2m+2)$ has height $(m+1)(y+m)/m$. This is $m+1>1$ at $y=0$, so $q_4=(0,1)$ lies below the line, and $(m+1)^2/m<2m+2$ at $y=1$, so $q_3=(1,2m+2)$ lies above it. Hence q_1q_5 is a diagonal and $q_1q_3q_5q_4$ is the cyclic order. (The non-vertex point $q_{13}=(-m,1-m,1)$ lies on plane (3) inside this quadrilateral.) Let $\Pi=\mathrm{conv}\{q_1,\dots,q_{12}\}$. Each of the sixteen planes contains at least three non-collinear points q_k of its list (the zero entries of Table 3), so it meets $\Pi \subseteq H$ in a 2-dimensional face, i.e. a facet of Π. These sixteen polygons (12 triangles and the 4 quadrilaterals $(2),(3),(2'),(3')$) have 26 edges, each shared by exactly two of them; since the facet-adjacency graph of a 3-polytope is connected, they are all the facets of Π, and $\Pi=H$. As $\Pi \subseteq \Phi(\mathcal{G}(S)) \subseteq H=\Pi$, the gem is the image of Π under Φ^{-1}, with vertex set the twelve forced vertices and f-vector $(12,26,16)$.

Step 5: isometries. S is associative (Step 1), so $\mathrm{Iso}(P_S)=\{\pm\mathrm{id}\}$ by Theorem 4.2. An isometry of $\mathcal{G}(S)$ fixes the centroid $\mathbf{0}$ of its centrally symmetric vertex set and is orthogonal. Write $\tilde{q}_k=\Phi^{-1}(q_k)$ for the forced vertices in the original coordinates: $\tilde{q}_1=(-m,m,m{+}2)$, $\tilde{q}_2=(-1,m,2m^2{+}2m-1)$, $\tilde{q}_3=(-m,1,2m^2+m)$, $\tilde{q}_4=(-m,0,-2m^2-m+1)$, $\tilde{q}_5=(-m,-m,-m)$, $\tilde{q}_6=(0,-m,2m^2{+}2m)$, and the other six are their negatives. The squared norms $3m^2<3m^2{+}4m{+}4<4m^4{+}4m^3{-}2m^2{-}2m{+}1<4m^4{+}4m^3{+}2m^2{+}1<4m^4{+}8m^3{+}m^2{-}4m{+}2<4m^4{+}8m^3{+}5m^2$ of $\tilde{q}_5,\tilde{q}_1,\tilde{q}_4,\tilde{q}_3,\tilde{q}_2,\tilde{q}_6$ are distinct for $m\ge 2$, so an isometry maps each $\tilde{q}_k$ to $\pm\tilde{q}_k$. The inner products of $\tilde{q}_1$ with $\tilde{q}_2,\dots,\tilde{q}_6$ are $2m^3{+}7m^2{+}4m{-}2$, $m(2m^2{+}6m{+}3)$, $-(2m^3{+}4m^2{+}m{-}2)$, $-m(m{+}2)$, $m(2m^2+5m+4)$, all non-zero, so all signs agree with that of $\tilde{q}_1$, and the isometry is $\pm\mathrm{id}$. □

All entries of Table 3 and the closure of the facet list are verified by `anc/siamese_slacks.py` in exact polynomial arithmetic. As a cross-check, not part of the proof, the theorem agrees with certified exact hull computations for $n=3,5,\dots,41$ (`anc/verify_structure.py`). For $m=1$ several entries vanish and the four triangles degenerate.

Table 3: Proof of Theorem 6.2, Step 4: the sixteen vertices $q_1, \dots, q_{16}$ of the level polygons $\Lambda_{+2}, \Lambda_{+1}, \Lambda_0, \Lambda_{-1}, \Lambda_{-2}$ (groups of rows) and the slacks of the inequalities (1)–(8) at them, as polynomials in m; zeros in bold. All non-zero entries are positive for $m \geq 2$. The points $q_1, \dots, q_{12}$ are the images of the forced vertices; $q_{13}, \dots, q_{16}$ are not vertices of the gem.

point	(x, y, ζ)	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
q_1	$(-m, m, 2m+2)$	**0**	**0**	**0**	**0**	$2(m-1)(m+1)$	$2m(m-1)(m+2)$	$2m(m-1)$	$2(m-1)$
q_2	$(-1, m, 2m+2)$	**0**	**0**	$m-1$	$2(m-1)(m+1)$	**0**	**0**	**0**	$2(m-1)$
q_3	$(-m, 1, 2m+2)$	**0**	$m-1$	**0**	$(m-1)(2m+1)$	**0**	$(m-1)(2m+1)$	$(m-1)(2m-1)$	**0**
q_4	$(-m, 0, 1)$	$2m+1$	m	**0**	**0**	$4m^2+2m-1$	$2m(2m^2+2m-1)$	$2m(2m-1)$	$2m-1$
q_{13}	$(-m, 1-m, 1)$	$2m+1$	$2m-1$	**0**	$(m-1)(2m+1)$	$2m^2+2m+1$	$(m+1)(2m^2+2m-1)$	$4m^2-3m+1$	1
q_{14}	$(-1, m-1, 1)$	$2m+1$	1	$m-1$	$m-1$	$4m^2+2m-1$	$(m+1)(4m^2-2m+1)$	$(m+1)(2m-1)$	$4m-3$
q_5	$(-m, -m, 0)$	$2(m+1)$	$2m$	**0**	$2m^2$	$2m(m+1)$	$2m^2(m+2)$	$2m(2m-1)$	**0**
q_6	$(0, -m, 0)$	$2(m+1)$	$2m$	m	$2m(2m+1)$	**0**	**0**	$2m(m-1)$	**0**
q_7	$(0, m, 0)$	$2(m+1)$	**0**	m	**0**	$4m(m+1)$	$2m(2m^2+2m-1)$	$2m^2$	$4m$
q_8	$(m, m, 0)$	$2(m+1)$	**0**	$2m$	$2m(m+1)$	$2m(m+1)$	$2m(m-1)(m+1)$	**0**	$4m$
q_9	$(m, 0, -1)$	$2m+3$	m	$2m$	$2m(2m+1)$	$2m+1$	**0**	**0**	$2m+1$
q_{15}	$(m, m-1, -1)$	$2m+3$	1	$2m$	$(m+1)(2m+1)$	$2m^2+2m-1$	$(m-1)(2m^2+2m-1)$	$m-1$	$4m-1$
q_{16}	$(1, 1-m, -1)$	$2m+3$	$2m-1$	$m+1$	$4m^2+m+1$	$2m+1$	$(m-1)(2m+1)$	$(m-1)(2m-1)$	3
q_{10}	$(m, -m, -2m-2)$	$4(m+1)$	$2m$	$2m$	$2m(2m+1)$	$2(m+1)^2$	$2m(m^2+m+1)$	$2m^2$	$2(m+1)$
q_{11}	$(m, -1, -2m-2)$	$4(m+1)$	$m+1$	$2m$	$(m+1)(2m+1)$	$4m(m+1)$	$(m+1)(4m^2-2m+1)$	$(m+1)(2m-1)$	$4m$
q_{12}	$(1, -m, -2m-2)$	$4(m+1)$	$2m$	$m+1$	$2(m^2+m+1)$	$4m(m+1)$	$2m(2m^2+2m-1)$	$2m(2m-1)$	$2(m+1)$

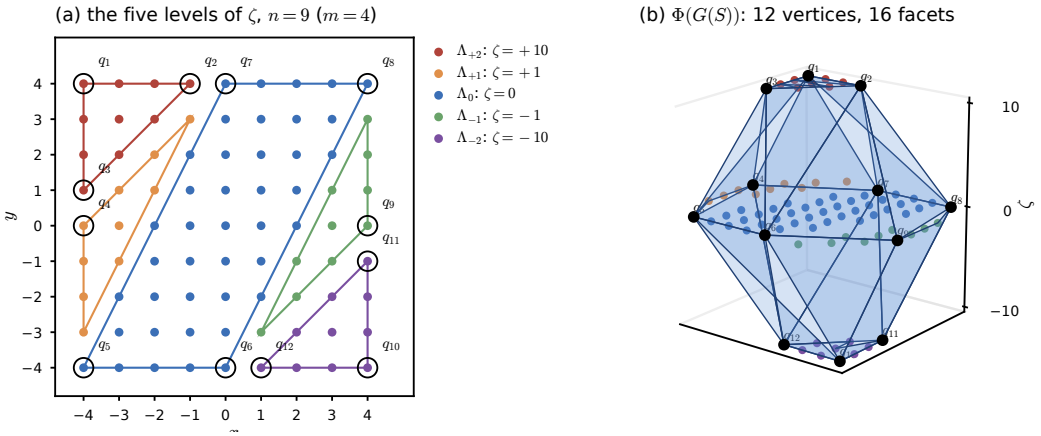


Figure 8: The Siamese gem of order $n = 9$ after the shear Φ. (a) The 81 cells coloured by the level value $\zeta \in \{\pm 10, \pm 1, 0\}$, with the five level polygons outlined and the twelve vertex cells $q_1, \dots, q_{12}$ circled. (b) $\Phi(\mathcal{G}(S))$ with its 12 vertices and 16 facets. The combinatorics of the level polygons does not depend on $m \geq 2$.

Remark 6.3 (Linear squares). The Siamese square is a *linear* (cyclic) square $s(c) = n\langle \mathbf{u} \cdot c + \alpha \rangle + \langle \mathbf{w} \cdot c + \beta \rangle + 1$ with $\mathbf{u}, \mathbf{w} \in \mathbb{Z}_n^2$ and $\alpha, \beta \in \mathbb{Z}_n$, and for fixed integer vectors $\mathbf{u}, \mathbf{w}$ the same shear reduces the lift to a number of levels that does not depend on n. The gems of linear squares do not share the twelve-vertex count, however. Over all $(\mathbf{u}, \mathbf{w}, \alpha, \beta) \in \mathbb{Z}_n^2 \times \mathbb{Z}_n^2 \times \mathbb{Z}_n \times \mathbb{Z}_n$ for which s is a magic square, counted up to D_4 (each family is closed under complementation; complements are not identified), the vertex counts (`anc/linear_squares.py`, exact hulls) range over $[11, 18]$ for $n = 5$ (184 D_4-classes), $[12, 25]$ for $n = 7$ (3159 classes), $[12, 31]$ for $n = 9$ (441 classes) and $[12, 36]$ for $n = 11$ (89,125 classes). For $n = 5$ the minimum 11 is attained by two classes, complements of each other, both pandiagonal and not associative, with f-vector $(11, 23, 14)$; one is $s = 5\langle 4i + 3j \rangle + \langle 4i + 2j \rangle + 1$ ($\mathbf{u} = (-1, -2)$, $\mathbf{w} = (-1, 2)$, $\alpha = \beta = 0$), the first square below.

$$\begin{pmatrix} 1 & 18 & 10 & 22 & 14 \\ 25 & 12 & 4 & 16 & 8 \\ 19 & 6 & 23 & 15 & 2 \\ 13 & 5 & 17 & 9 & 21 \\ 7 & 24 & 11 & 3 & 20 \end{pmatrix} \qquad \begin{pmatrix} 1 & 19 & 15 & 13 & 17 \\ 23 & 5 & 14 & 20 & 3 \\ 10 & 21 & 12 & 4 & 18 \\ 24 & 9 & 8 & 22 & 2 \\ 7 & 11 & 16 & 6 & 25 \end{pmatrix}$$

For $n = 7, 9, 11$ the minimum 12 is attained by exactly two classes, one of them the Siamese square and the other the digit-exchanged square s' of Lemma 6.1; its gem has f-vector $(12, 24, 14)$ for $n = 5, 7, \dots, 31$. The Siamese gem is not vertex-minimal at order five: simulated annealing (Appendix B) found 15 D_4-classes of magic squares of order 5 whose gems have 10 vertices

(`anc/sa_n5_V10.txt`). For example, the second square above has f-vector $(10, 22, 14)$, volume $845/3$ and 8 interior points.

7 Open questions

1. *Hull versus point set.* Is $\mathrm{Iso}(\mathcal{G}(S)) = \mathrm{Iso}(P_S)$ for every magic, or semi-magic, square S? By Theorem 4.2 this would determine the isometry groups of all gems; it holds for order four (Theorem 4.3) and for the Siamese squares (Theorem 6.2). For odd n the self-complementary squares are associative (Lemma 4.1), so the point set of an order-five square has isometry group $\{\pm\mathrm{id}\}$ or trivial; whether the same holds for the gems is open.
2. *Order five.* Schroeppel's enumeration gives 275,305,224 magic squares of order five up to D_4 [14]. At the rate of the order-four computation (about 16 ms per certified exact hull) a full census would take about seven CPU-weeks. How many combinatorial types and congruence classes occur, and are the congruence classes again the complement orbits?
3. *Growth of the vertex count.* For uniformly random arrangements the mean vertex count grows from 8.22 at $n = 3$ to 27.91 at $n = 10$, and the ratio $\mathbb{E}[V]/\ln^2 N$ decreases slowly from 1.70 to 1.32; samples of magic squares obtained by simulated annealing have means within 1.2 standard errors of the random ones for $n = 5, \ldots, 8$ (Table 4; Monte Carlo estimates, not results). For N independent uniform points in a 3-polytope the expected vertex count is $\Theta(\ln^2 N)$ [3]; our points are not independent (the grid is fixed and only the heights are permuted), so this is only a heuristic analogy. Is $\mathbb{E}[V] = \Theta(\ln^2 n)$ for random arrangements, and for uniformly random magic squares?
4. *Secondary cones.* Which pairs of secondary cones of the $n \times n$ grid contain $(z, -z)$ for the height function z of a magic square (Proposition 5.2)? For $n = 4$ the 7040 magic height functions induce 417 regular subdivisions up to D_4.
5. *Linear squares.* Which linear squares (Remark 6.3) have a vertex count bounded independently of n, and does the digit-exchanged square s' of Lemma 6.1 have f-vector $(12, 24, 14)$ for every odd $n \geq 5$?
6. *Other families.* The construction applies to semi-magic, pandiagonal and Franklin squares [18], and Proposition 2.3 and Theorem 4.2 already cover the semi-magic case.

Use of AI systems

The author conceived the Magic Gem construction in a spoken conversation with a chatbot (OpenAI's ChatGPT), used as a sounding board; the transcript is available from the author. All versions of this paper were prepared with substantial help from generative AI, which earlier versions did not state. Versions 1–4 were prepared with a mix of OpenAI's ChatGPT, Anthropic's Claude (Sonnet and Opus, used in the Cursor editor) and Google's Gemini, used for discussion, code and tests over the three years of the project. This version was restructured and largely rewritten with Anthropic's Claude, used through the agentic tool Claude Code in October 2026, which wrote and ran the code in `anc/`, proposed and wrote out proofs, and drafted the text; separate AI sessions acted as adversarial referees. Several errors in earlier versions were found in that process and are corrected here. The author directed the work and is responsible for the content; AI systems are not authors. Every number in the paper is recomputed by the scripts in `anc/`, the order-4 results by exact enumeration of all 880 squares. **[Author to confirm or edit before submission: which proofs the author has checked line by line.]**

Changes from earlier versions. Versions 1–4 contained the following errors and overstatements, which are corrected or withdrawn here. (i) The "Complete Energy Characterization Theorem" presented in version 4 as the main theoretical contribution is immediate from the definitions:

the energy is a sum of squared line-sum deviations, so it vanishes exactly on magic squares. It is not presented as a result here. (ii) Table 2 of version 4 gave about 3.4×10^7 essentially different magic squares of order five; the correct number is 275,305,224 [14], and 3.4×10^7 is that number divided by 8. (iii) The same table gave 8, 12 and 16 as the "typical" numbers of hull vertices for orders 3, 4 and 5. For order four the number ranges from 8 to 16 (Table 2 of this version), and the Siamese gem of order five has 12 vertices (Theorem 6.2). (iv) The energy-landscape statistics of version 4 (its Section 4.5, including the "landscape entropy" and the claim that it is scale-invariant) are not carried over; exact results on random arrangements are in the companion paper [20]. (v) The statistics in Table 3 of version 4 weighted the diagonal terms of the energy differently from its stated definition. That table also reported that 6×10^7 uniformly random arrangements of order four contained no non-magic zero of the four-term energy; since 14,639,440 of the 16! arrangements are such zeros [20], about 42 were to be expected, and a rerun with the same number of samples finds 45 (`anc/v4_sampling_recheck.py`). That statement of version 4 is withdrawn. (vi) The hardware stated in version 4 was incorrect; the computations were run on an Apple M4 Max.

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A Computational methods

All computations use Python 3.9 with NumPy 2.0, SciPy 1.13 (Quickhull), SymPy 1.13 and NetworkX 3.2; the scripts and data are the ancillary files in `anc/`, whose `README` lists the commands.

Exact hulls. Doubling all coordinates makes every point an integer vector. A floating-point Quickhull triangulation is used only to propose candidate facet planes, which are recomputed as primitive integer equations from three points each; coplanar triangles are merged. The hull is then certified in integer arithmetic: every one of the N points satisfies every facet inequality, every edge of every facet polygon lies in exactly two facets (so the facets form a closed surface and are all the facets), and $V - E + F = 2$. This certificate is applied to every exact hull in the paper (order four, Siamese, digit-exchanged and linear squares, and the 15 ten-vertex order-five squares). Vertices are the points on at least three facet planes, interior points are those satisfying every facet inequality strictly, and volumes are exact rationals. For the 880 order-four gems this takes about 15 s in total.

Combinatorial types, isometries, congruence. Combinatorial types are the isomorphism classes of the bipartite vertex–facet incidence graphs, under isomorphisms that map vertices to vertices and facets to facets (so a polytope and its dual are not identified); candidates are grouped by a Weisfeiler–Lehman hash and confirmed by explicit isomorphism tests. Isometry groups of a finite set X (a vertex set or a full point set) are computed exactly: X is centred at its centroid, every isometry is then orthogonal and determined by the images of three linearly independent points, which are searched among the points with the same squared norms and mutual inner products, and a candidate is accepted only if the induced permutation of X preserves all pairwise integer squared distances. Two finite sets are congruent if and only if there is a bijection between them preserving all pairwise distances; this is tested by grouping by the sorted multiset of squared distances and then by an isomorphism test of the complete graphs with squared distances as edge labels. For the 880 order-four gems the 616 distinct distance multisets already certify the congruence classes (Proposition 5.4); the all-pairs test is a redundant check. The point-set statements of Theorem 4.2 are checked in the same way for all 880 squares and for the Siamese squares of orders 3 to 15.

Polynomial identities. The slack polynomials of Table 3 are computed with SymPy; a polynomial p is certified positive for all integers $m \geq 2$ by non-negativity of all coefficients of $p(2+t)$ and $p(2) > 0$. The same script checks the closure of the facet list and the norms and inner products of Step 5.

B Lattice-point counts and vertex statistics

Ehrhart polynomials of the order-four gems. For even n all coordinates of every $\mathbf{p}_{ij}$ are half-odd integers, so $Q(S) = \mathcal{G}(S) + (\frac{1}{2}, \frac{1}{2}, \frac{1}{2})$ is a lattice polytope; any other lattice translate has the same Ehrhart polynomial $L_S(t) = \#(tQ \cap \mathbb{Z}^3) = \mathrm{vol}(\mathcal{G})\, t^3 + c_2 t^2 + c_1 t + 1$ [4], and L_S is the same for all images gS, $g \in D_4$, since $\hat{g}$ preserves $\mathbb{Z}^3$ up to an integer translation of Q. Writing $\sum_{k \geq 0} L_S(k) t^k = (h_0^* + h_1^* t + h_2^* t^2 + h_3^* t^3)/(1-t)^4$, the h_i^* are non-negative integers [32], $h_0^* = 1$,

Table 4: Mean number of vertices of the gem. Random: uniformly random arrangements, 10^5 samples per order (standard errors at most 0.010). Magic: exact for $n = 3, 4$; for $n = 5, \ldots, 8$, annealing samples of 59,993, 24,000, 8,000 and 2,400 distinct D_4-classes (standard errors 0.007, 0.013, 0.025, 0.054). Siamese: exact.

n	3	4	5	6	7	8	9	10
random	8.22	12.06	15.39	18.39	21.08	23.53	25.79	27.91
magic	8	12.23	15.38	18.41	21.08	23.56	–	–
Siamese	8	–	12	–	12	–	12	–

$\sum h_i^* = 6\,\mathrm{vol}$, and h_3^* is the number of interior lattice points of Q [4]. We computed L_S exactly for all 880 gems (`anc/ehrhart_880.py`): the lattice points of tQ are counted for $t \le 6$, the cubic is interpolated from $t \le 3$ and checked at $t = 4, 5, 6$, h_3^* is checked against the number of interior lattice points and $\sum h_i^*$ against $6\,\mathrm{vol}$. All h^*-vectors are non-negative, with $h_1^* \in [70, 136]$, $h_2^* \in [233, 403]$ and $h_3^* \in [34, 52]$. There are 554 distinct Ehrhart polynomials, against 616 congruence classes, 352 combinatorial types, 39 f-vectors and 155 volumes. Congruent gems have equal polynomials (for complement partners, the reflection $z \mapsto -z$ followed by an integer translation is a unimodular affine map of $\mathbb{Z}^3$), but non-congruent gems can share one, and L_S is determined neither by the f-vector (it takes several values on 29 of the 39 f-vectors), nor by the combinatorial type (81 of 352 types), nor by the volume (104 of 155 volumes). For the Lo Shu square (odd order: the gem itself is a lattice polytope) $L(t) = 1 + t + 13t^2 + 18t^3$ and $h^* = (1, 29, 73, 5)$; for Dürer's square $L(t) = 1 - 3t + 44t^2 + 92t^3$ and $h^* = (1, 130, 377, 44)$; the leading coefficients are the volumes 18 and 92.

Vertex counts for larger orders (numerical). Table 4 lists the mean vertex counts of the gems of uniformly random arrangements, of magic squares (for $n = 5, \ldots, 8$ found by simulated annealing on the sum of squared line excesses, `anc/sa_magic.c`) and of the Siamese squares. For random arrangements the fraction $\mathbb{E}[V]/N$ falls from 0.91 ($n = 3$) to 0.28 ($n = 10$). At $n = 4$ magic gems have slightly more vertices than random ones; for $n = 5, \ldots, 8$ the annealed means are within 1.2 standard errors of the random ones. These are statements about the samples only: the sampling distribution of the annealer for $n \ge 5$ is not known (compare [26]). The smallest annealed vertex counts are 10, 12, 14 and 16 for $n = 5, \ldots, 8$ (Remark 6.3). The hulls of the samples are computed by Qhull in floating point, with the vertex sets of the first 100 hulls of each batch rechecked in exact arithmetic; the seeds are $7000n + c$ for random arrangements (`anc/vertex_statistics.py`) and $1000s + n$, $s = 1, \ldots, 16$, for the annealer (`anc/run_sa.sh`).